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FUTURE X FUTURE

1

EST. 2026

DESIGN

PORTFOLIO

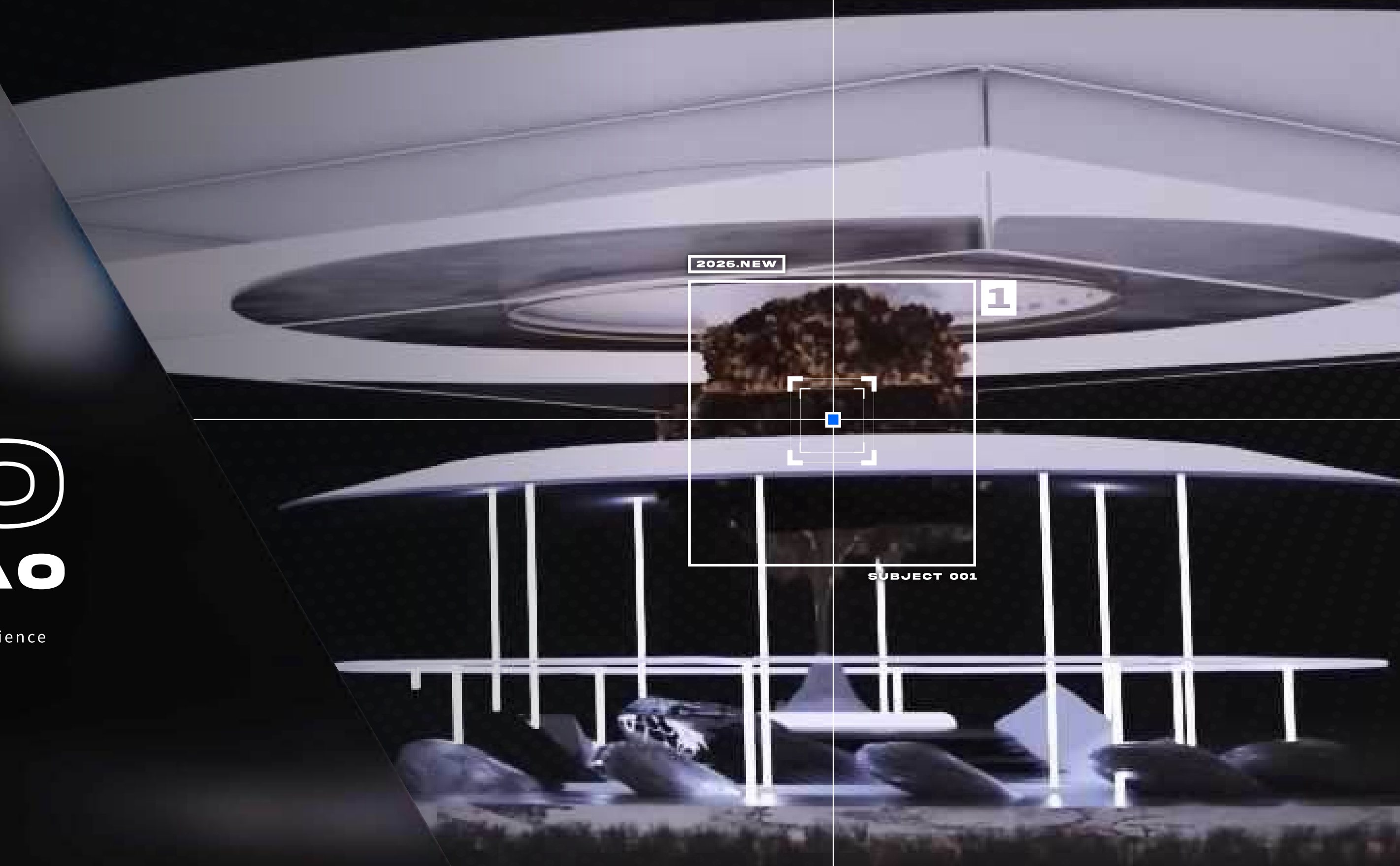
XIAN-HAO (HARRY) LIAO

Harry Liao excels at bridging creative strategy with sci-fi technology, possessing extensive experience in UX design, visual design, 3D modeling, motion graphics, and web development.

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DESIGN PORTFOLIO

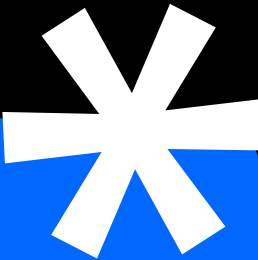
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DESIGN PORTFOLIO

INTRO

Hello,
I am Xian-Hao Liao 廖先皓, a UX designer with
a passion for futuristic aesthetics, storytelling,
and immersive experiences.



EXPERIENCE

2025 – 2026	GATE.COM
2025	BOSTON UNIVERSITY
2024-2025	THE UPS STORE
2023	BJ'S CLUB (INTERN)
2022 – 2024	MASSART
2021 – 2022	ABC DESIGN
2016 – 2020	NTCUST



SKILL

Adobe Illustrator
Adobe Photoshop
Adobe After Effects
Figma
Cinema 4D
HTML & CSS & JS
Nano Banana Pro



CONTACT

HOMEPAGE
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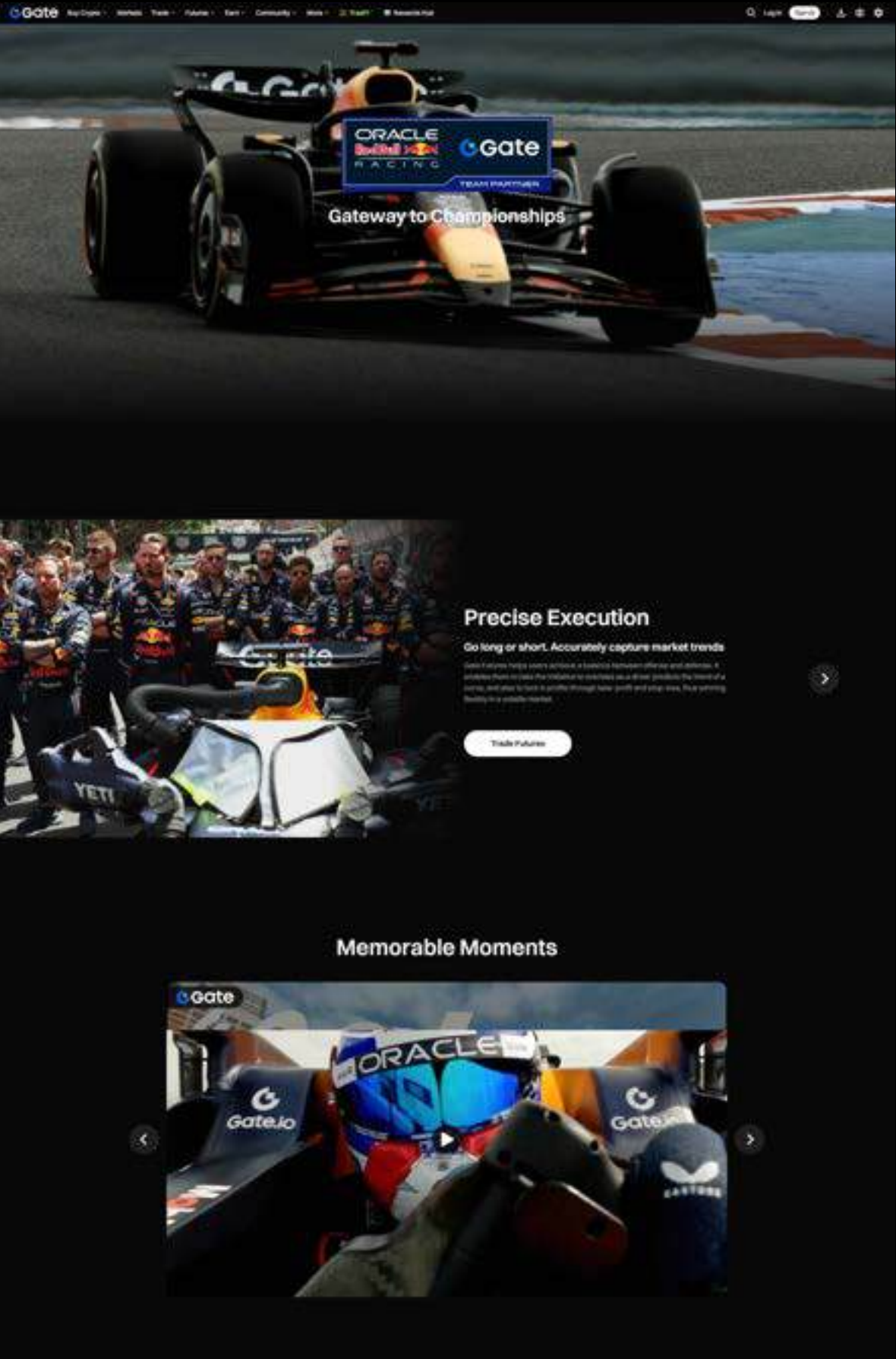
Gate



- Executed experience-focused UX design for the [Red Bull x Gate partners](#) page and events pages, ensuring high delivery standards.
- Accomplished social motion graphics for four high-quality [ad campaigns](#).

PROJECT SCOPE
UX/UI DESIGN SOCIAL MEDIA VISUALS MOTION GRAPHICS ICON DESIGN

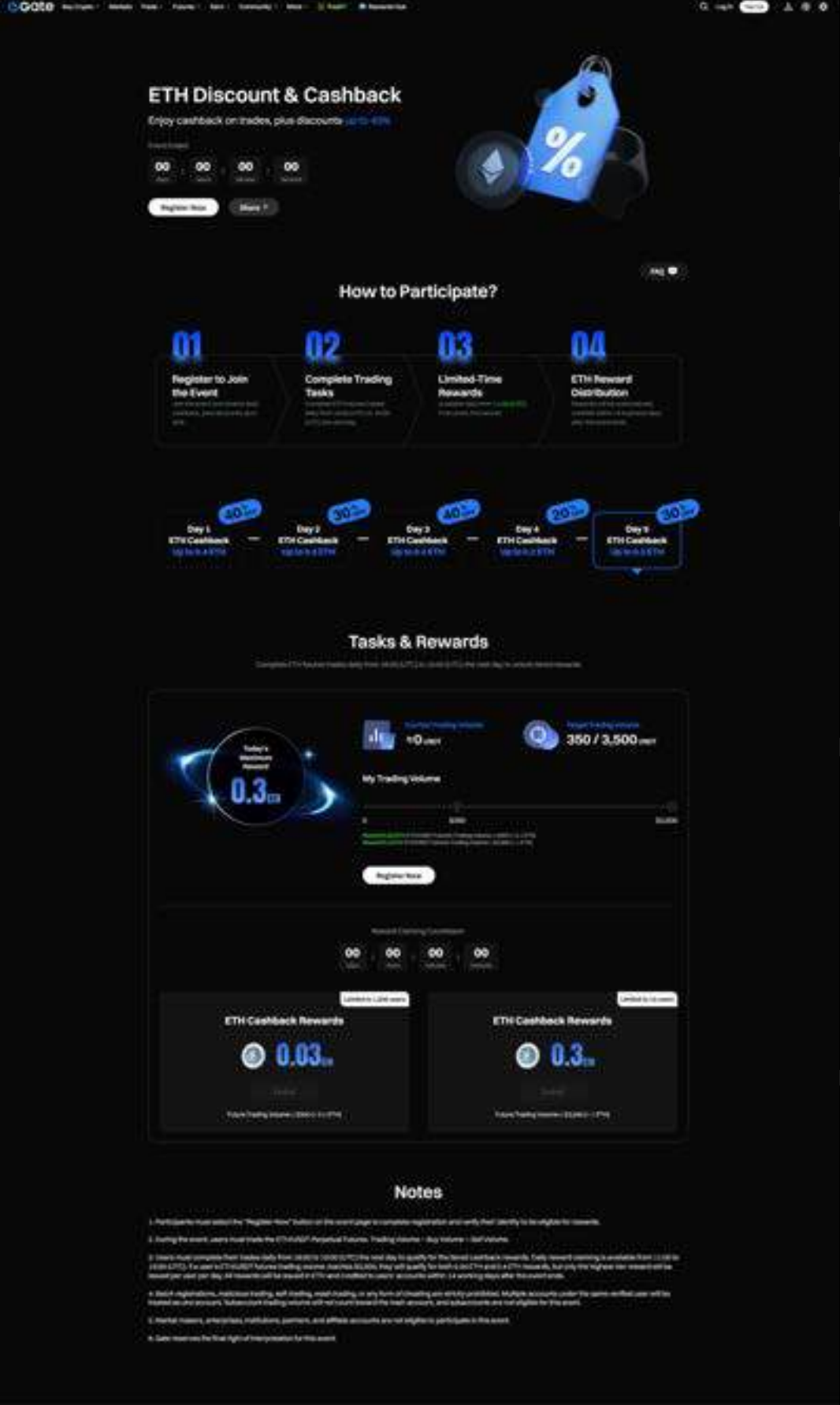
‘Gate x Redbull Racing’ Partners Page Design



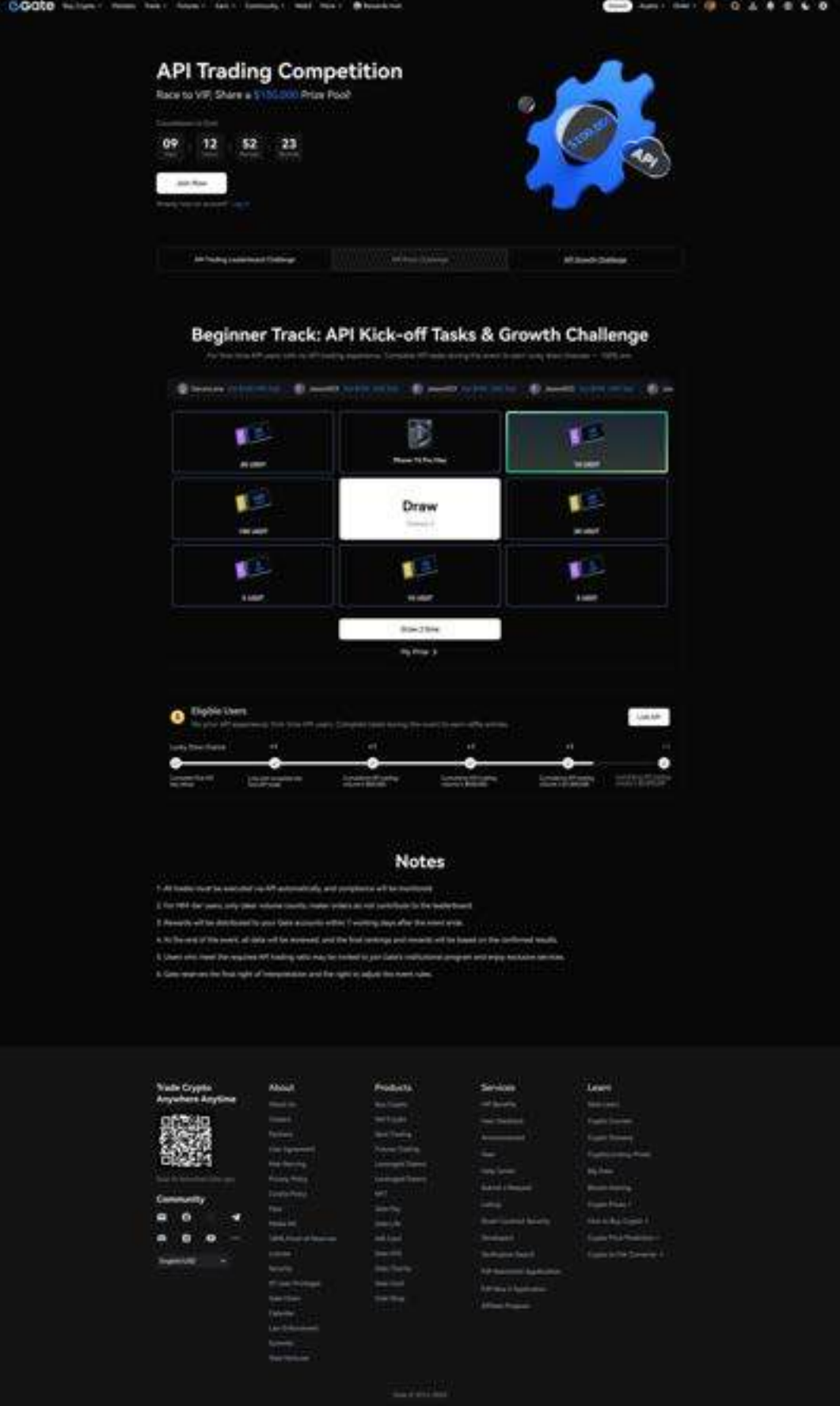
‘Gate x Redbull Racing’ NFT Website Design



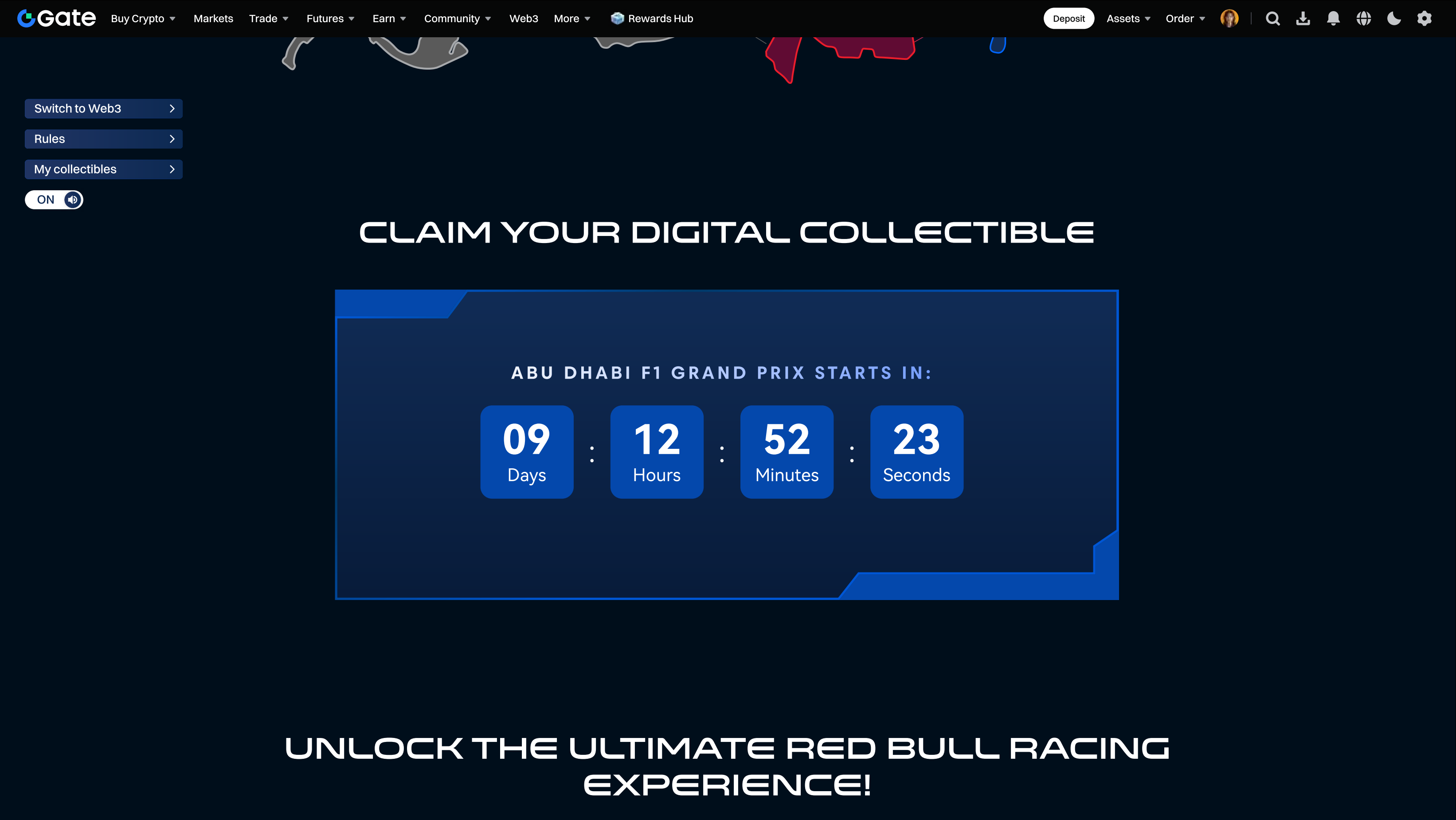
ETH Event Page Design



API Trading Page Design



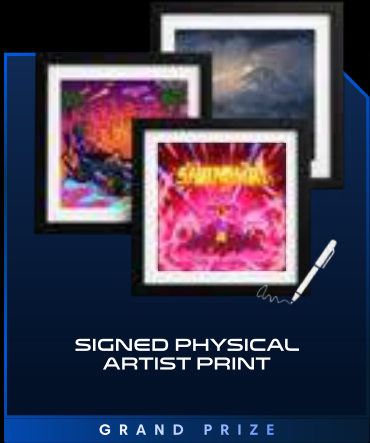
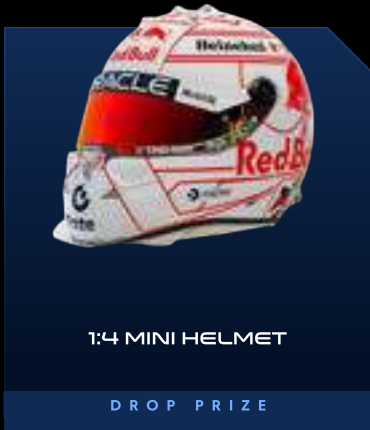
EVENT Website Design



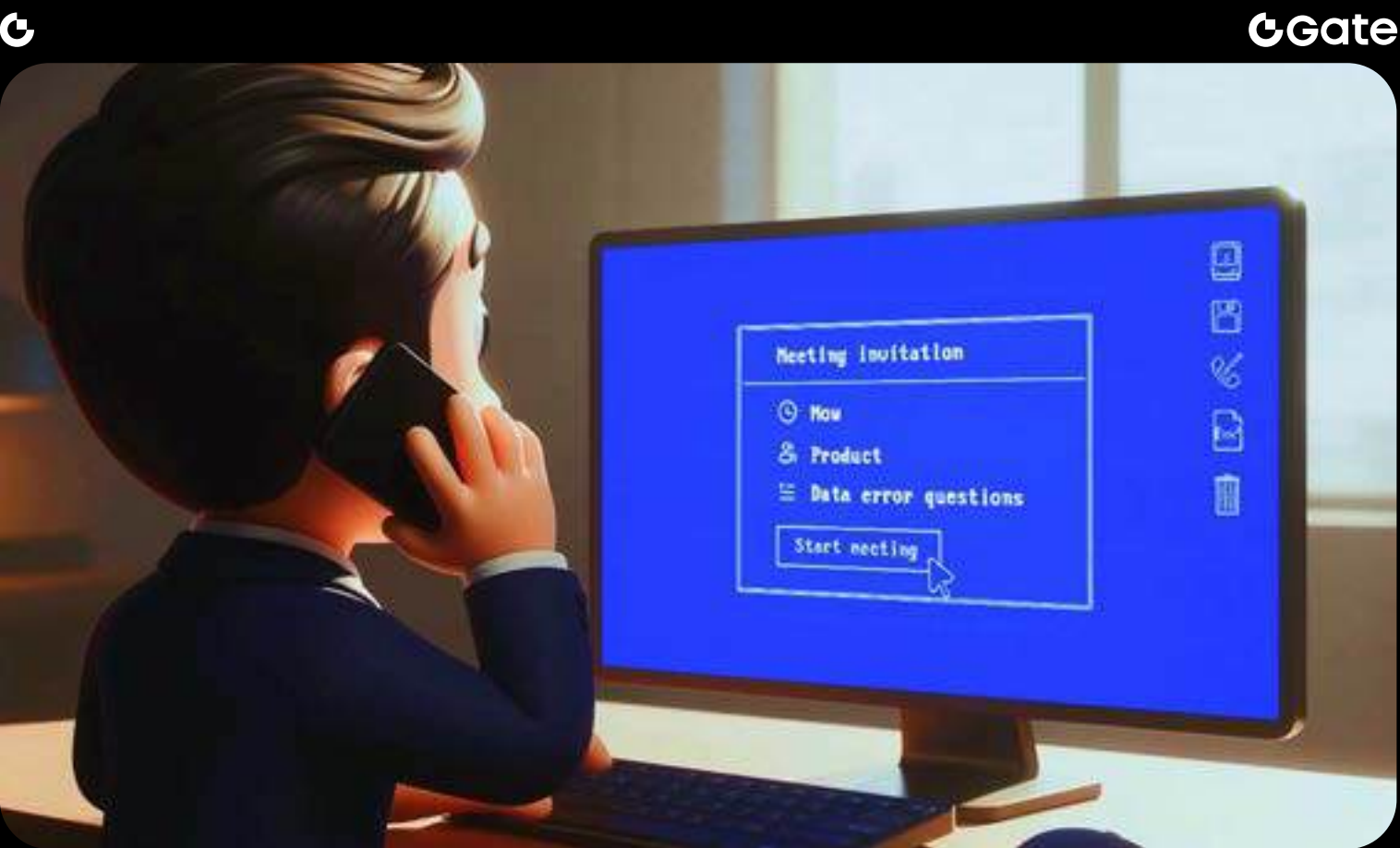
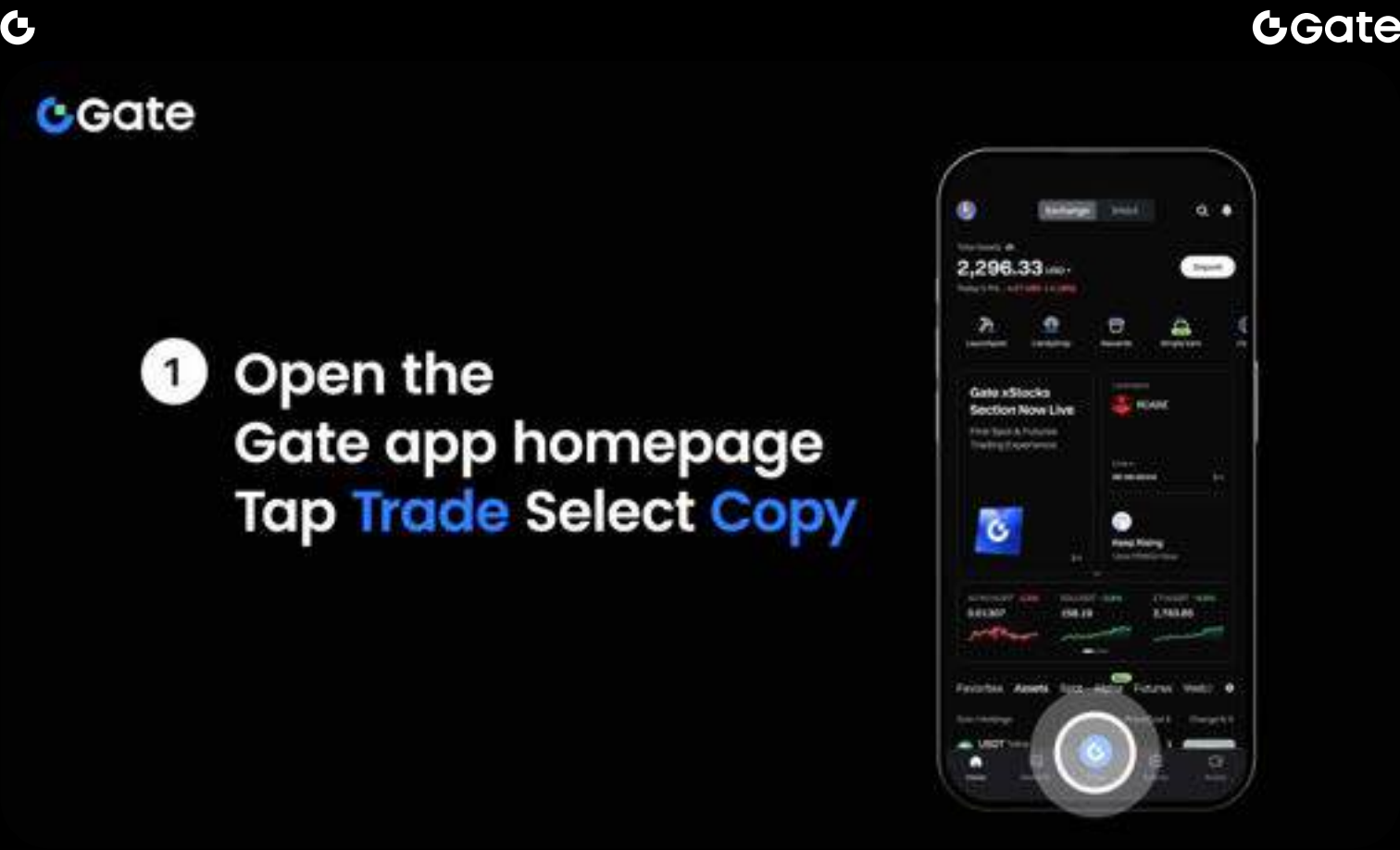
PROJECT TYPE
UX/UI DESIGN



PROJECT SCOPE
GATE X REDBULL RACING NFT EVENT WEBSITE DESIGN



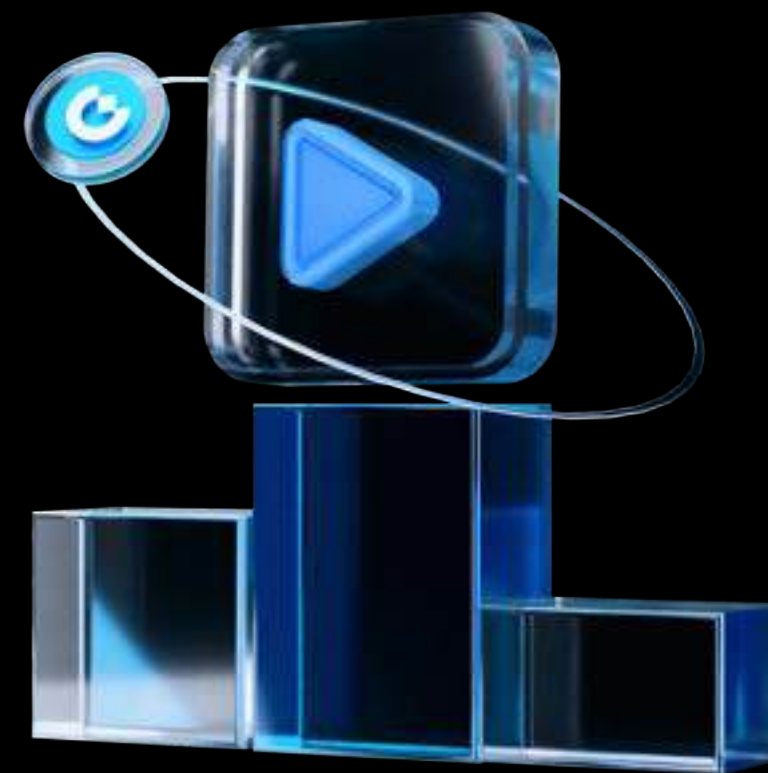
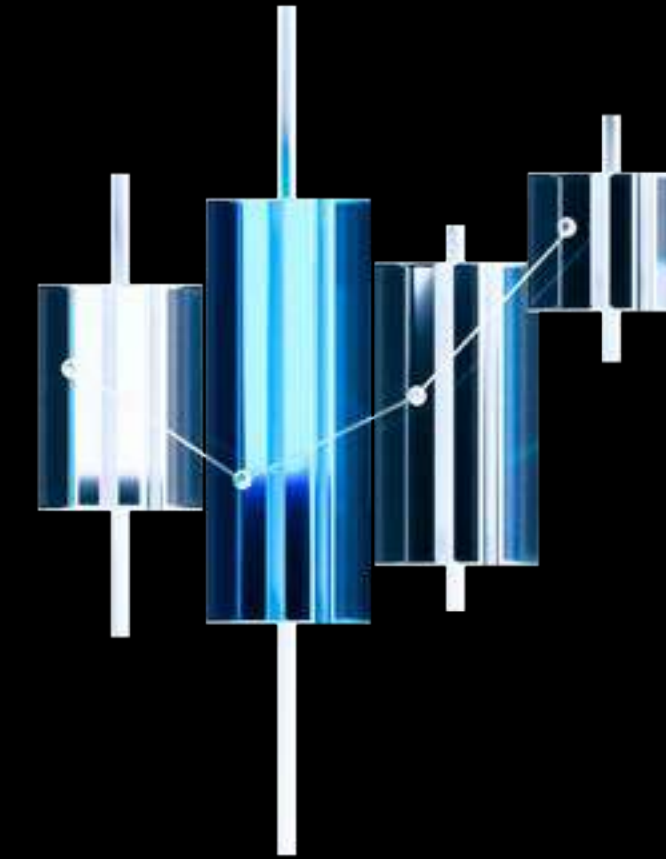
Motion Graphic Design



GATE

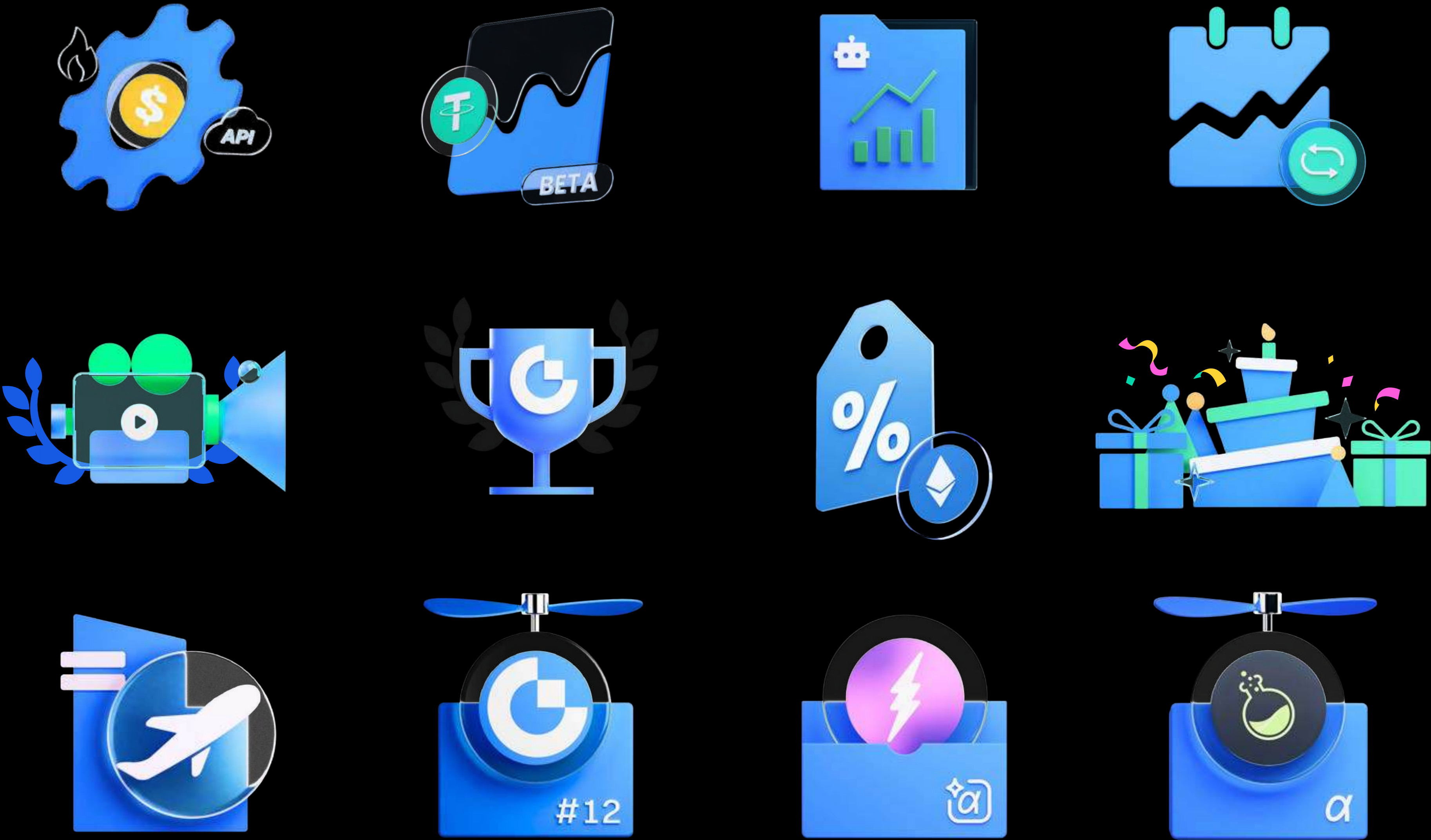
3D ICON DESIGN

Collaborated closely with the team to craft dynamic 3D icons for Gate.com event campaigns. These designs powered the V3.0 UI ecosystem prior to the system upgrades.



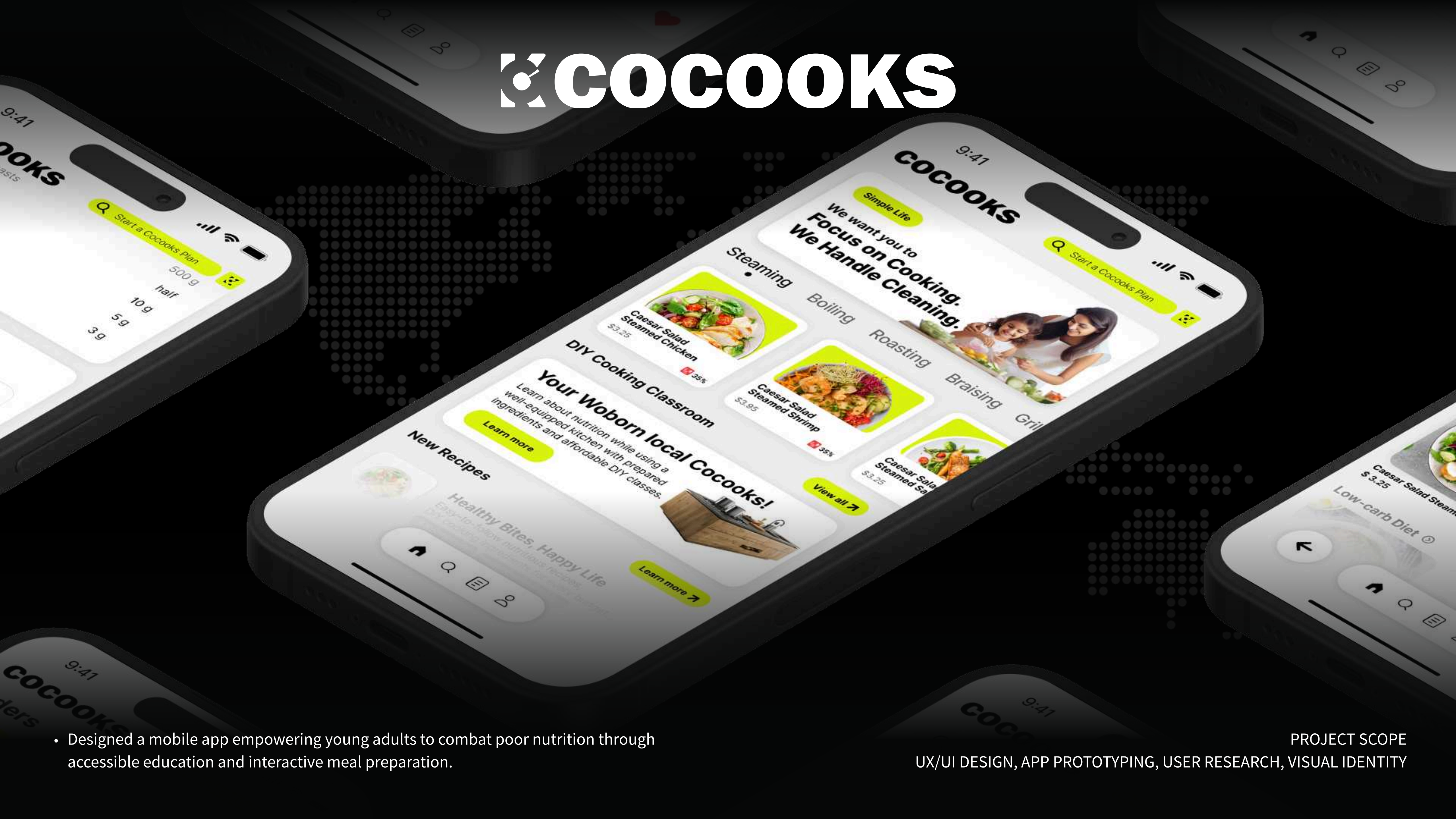
GATE
EVENT ICON DESIGN

In collaboration with the team, I designed custom 3D icons tailored for various Web3 event pages.



- EVENT ICON SYSTEM LIST
- API TRADING EVENT
 - CALL/PUT OPTION BETA
 - TRADING BOT REPORT
 - ROLLING SHORT OPTIONS
 - REWARDS & ACHIEVEMENTS
 - CRYPTO DISCOUNT EVENT
 - ANNIVERSARY CELEBRATIONS
 - TRAVEL & GLOBAL ECOSYSTEM
 - CRYPTO WALLETS & ASSETS

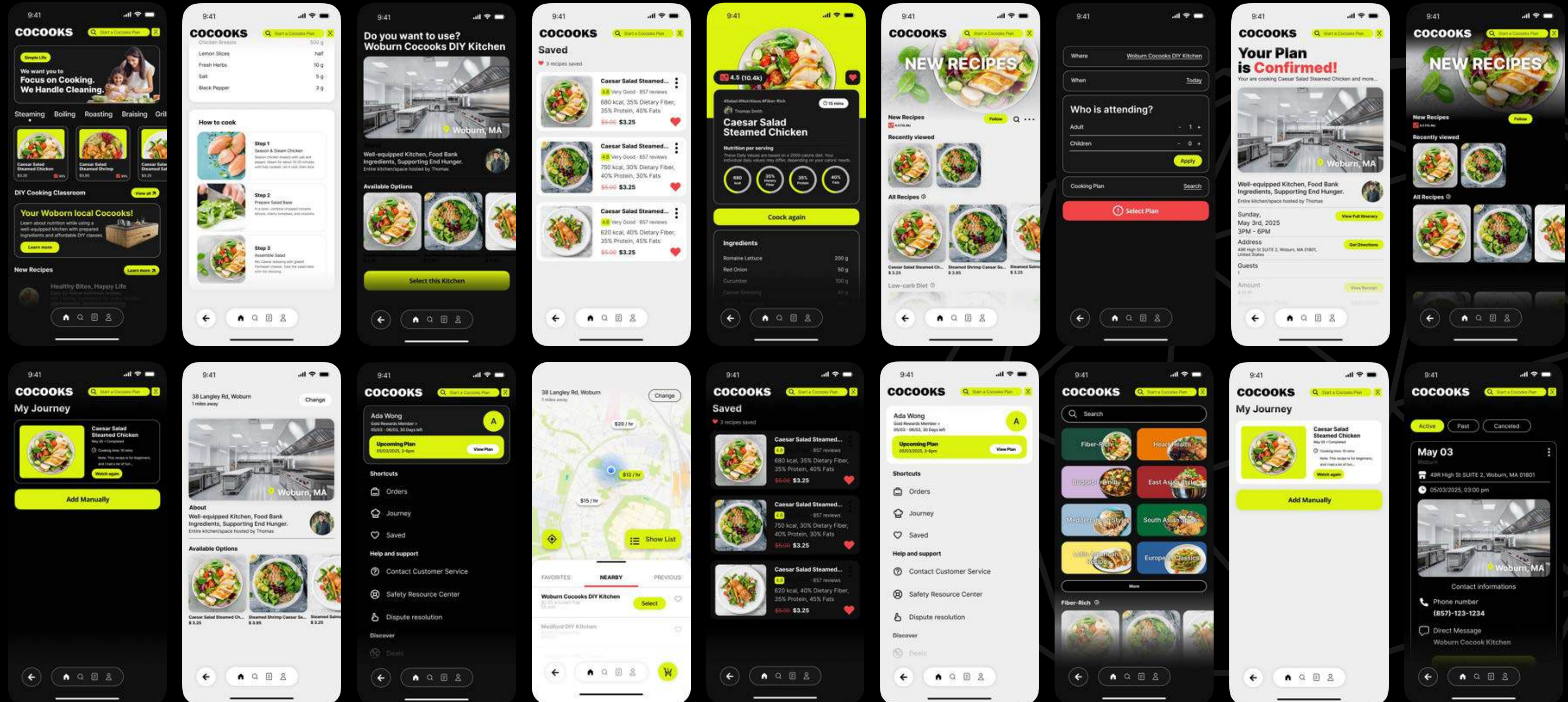
COCOOKS



- Designed a mobile app empowering young adults to combat poor nutrition through accessible education and interactive meal preparation.

PROJECT SCOPE
UX/UI DESIGN, APP PROTOTYPING, USER RESEARCH, VISUAL IDENTITY

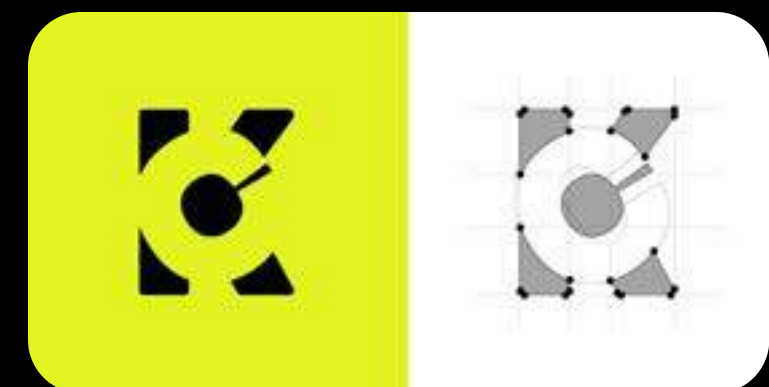
App UX Design



COCOOKS

INTEGRATED SERVICE DESIGN

Designed a holistic ecosystem bridging the digital and physical realms. By integrating an intuitive mobile app with shared DIY cooking spaces, Cocooks empowers young adults to seamlessly embrace healthy and affordable meal preparation.



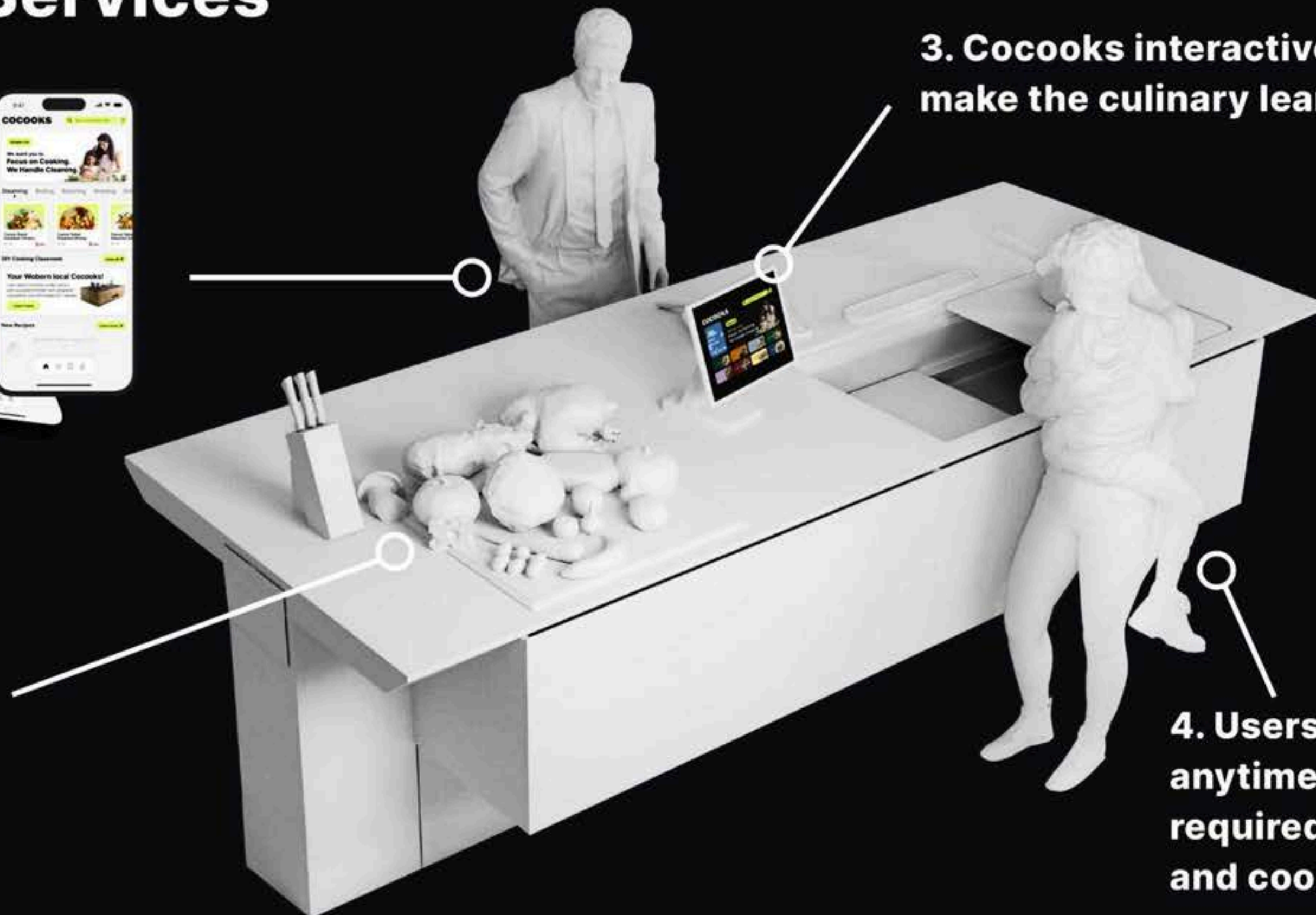
PROJECT TYPE
UX/UI DESIGN

Cocooks, An Affordable DIY Cooking Solution That Integrates Services

1. Foodies uses our **mobile app** to book cooking classes.



2. Cocooks provide space and cooking kits.



3. Cocooks interactive tutorials make the culinary learnings easy.

4. Users can drop in anytime—no planning required, just come and cook right away!

COCOOKS ISOMETRIC ILLUSTRATION

Designed a series of isometric line-art illustrations to visualize the fresh ingredient supply chain and urban delivery system for the Cocooks platform.

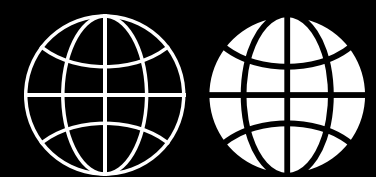
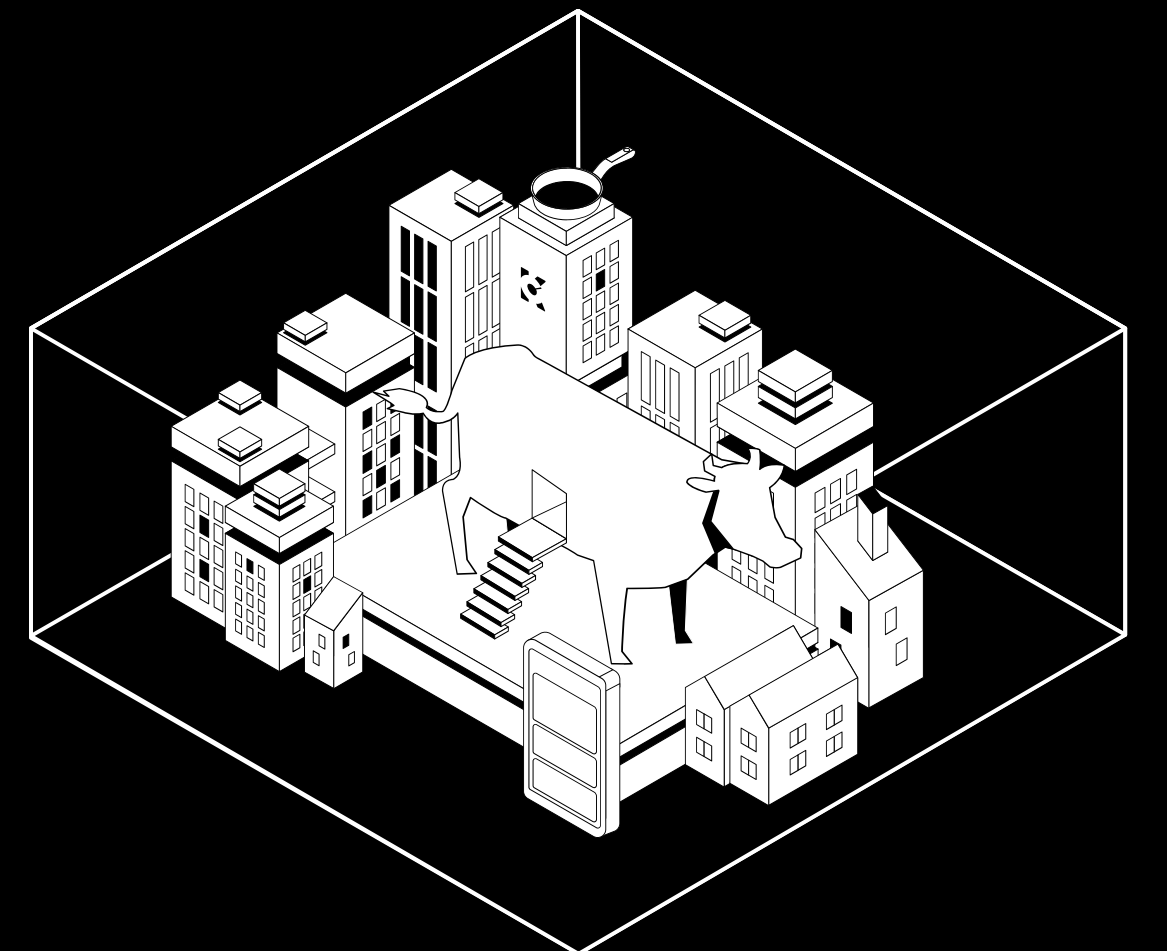
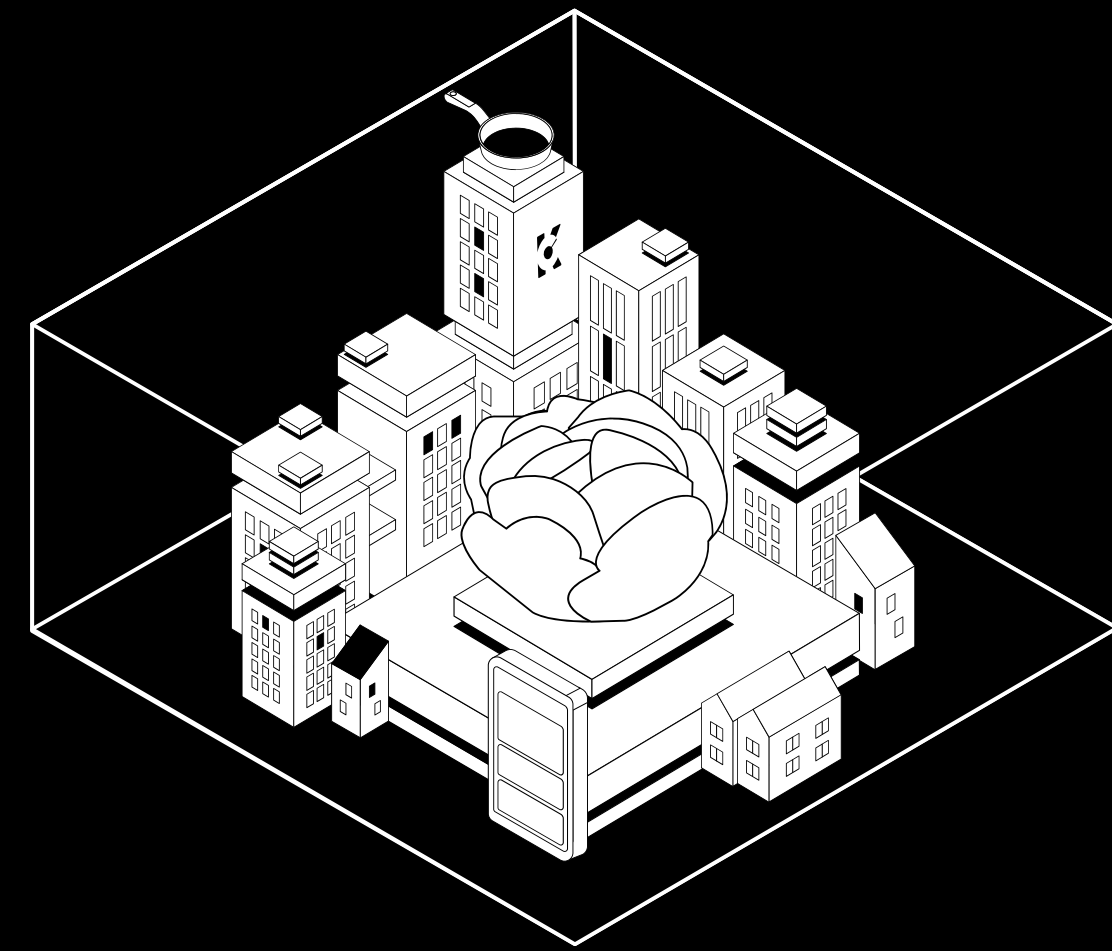
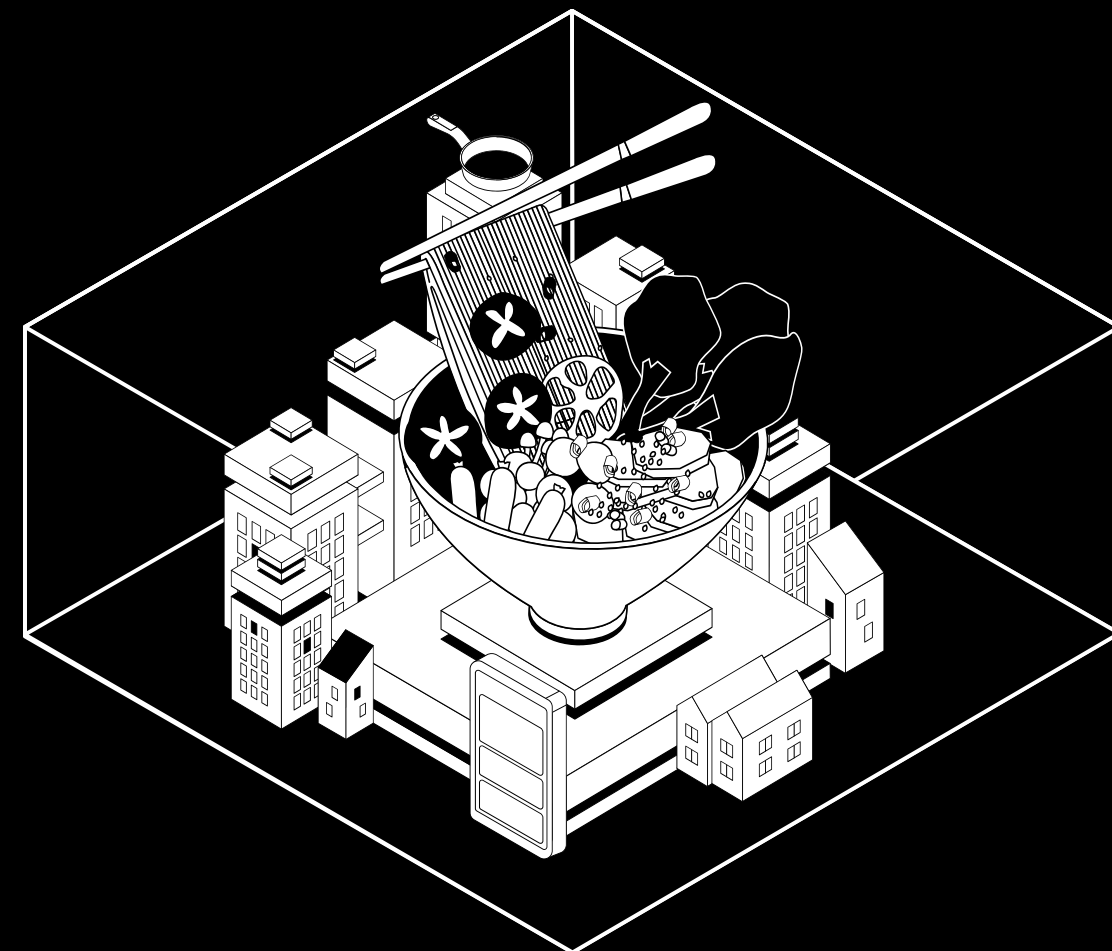
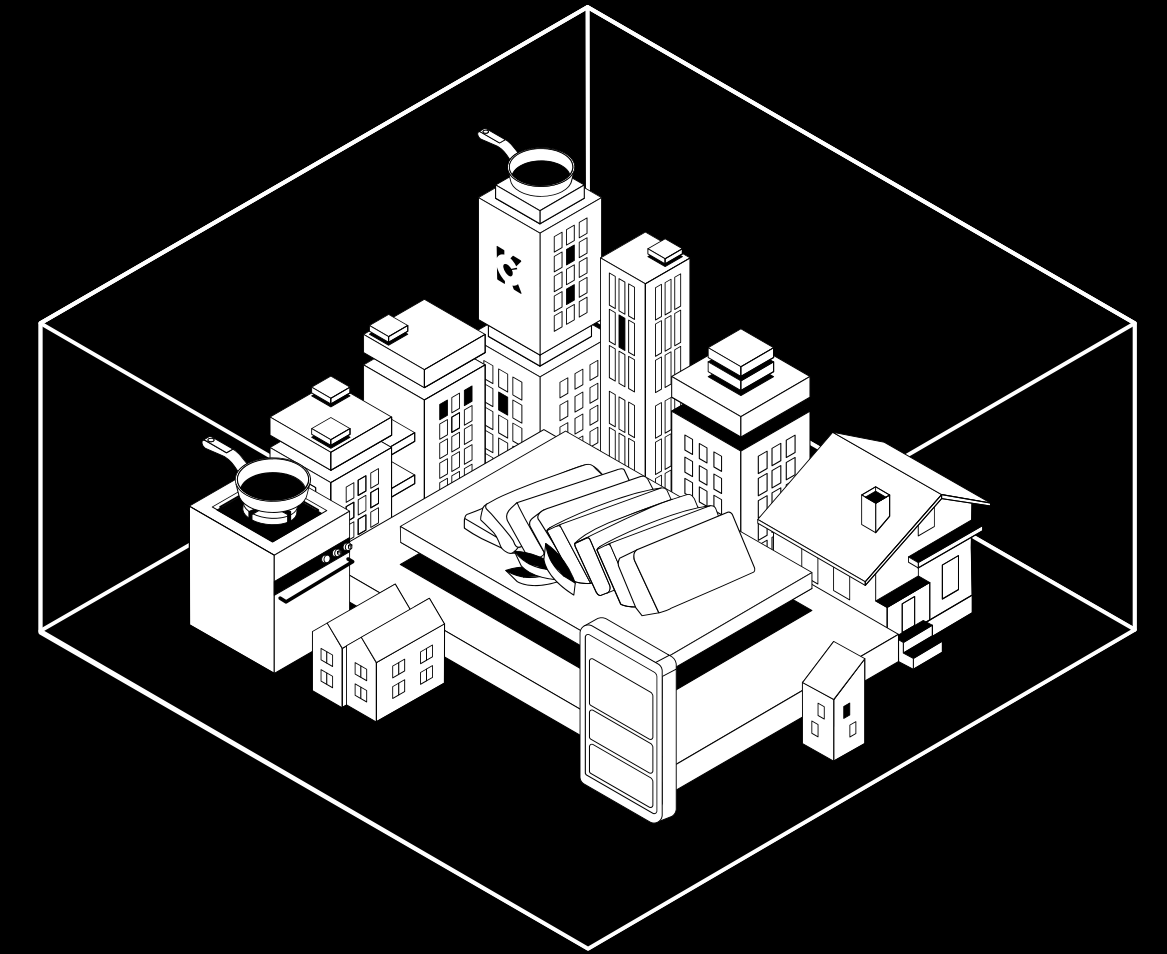
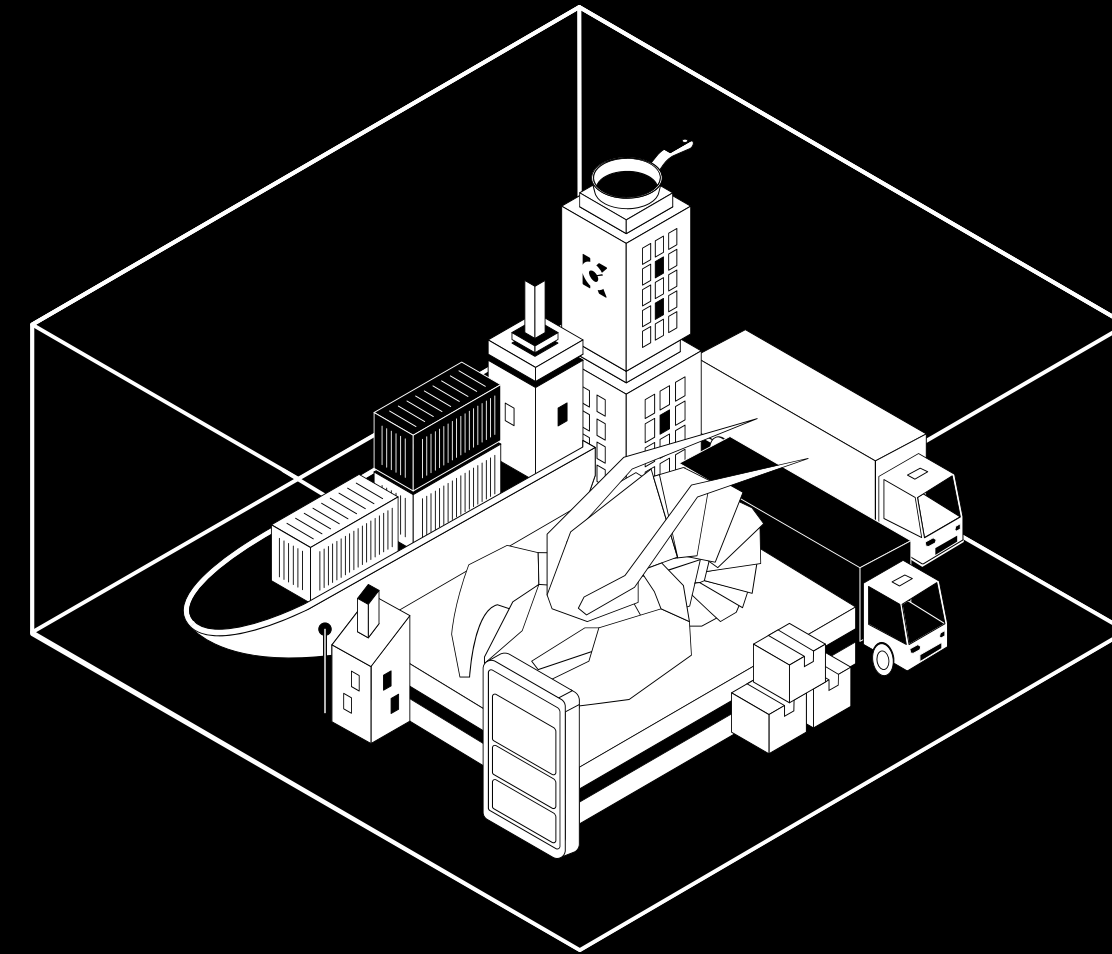
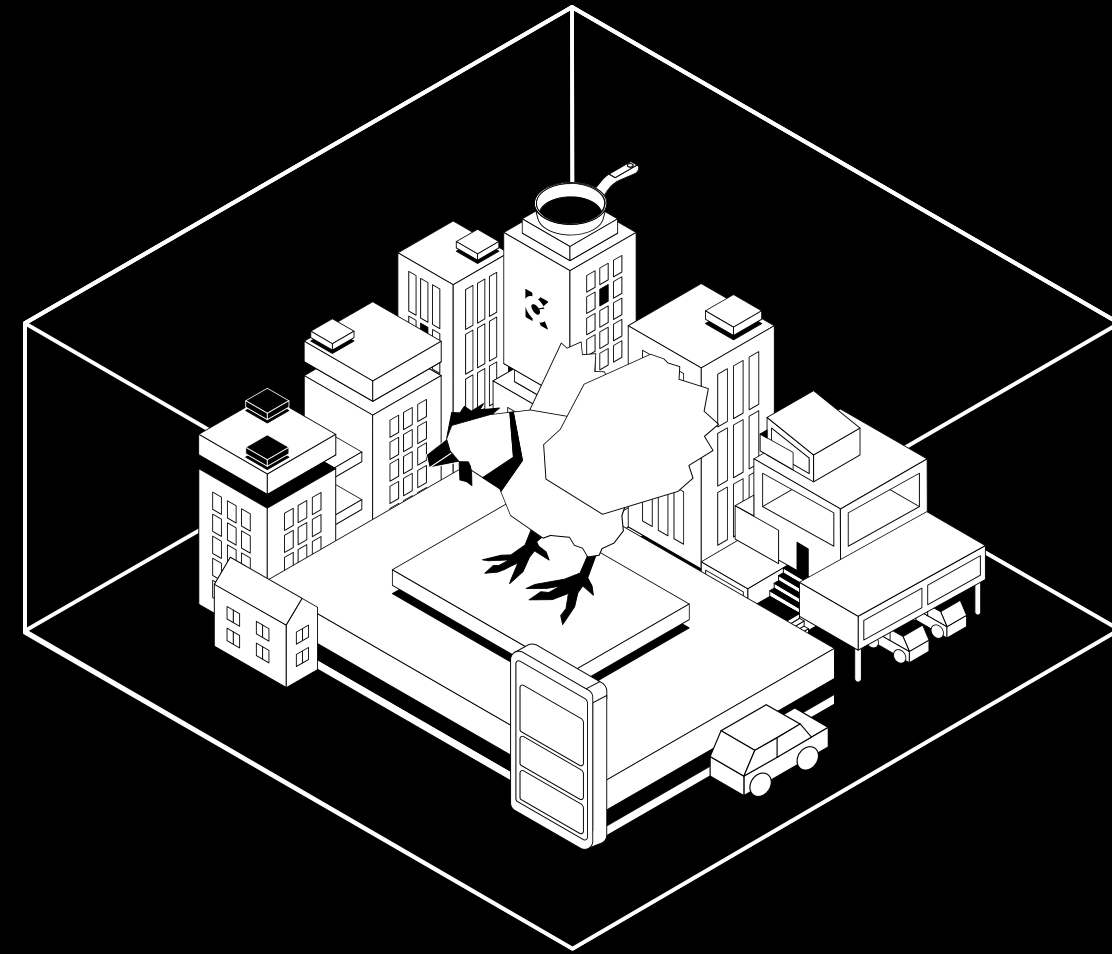
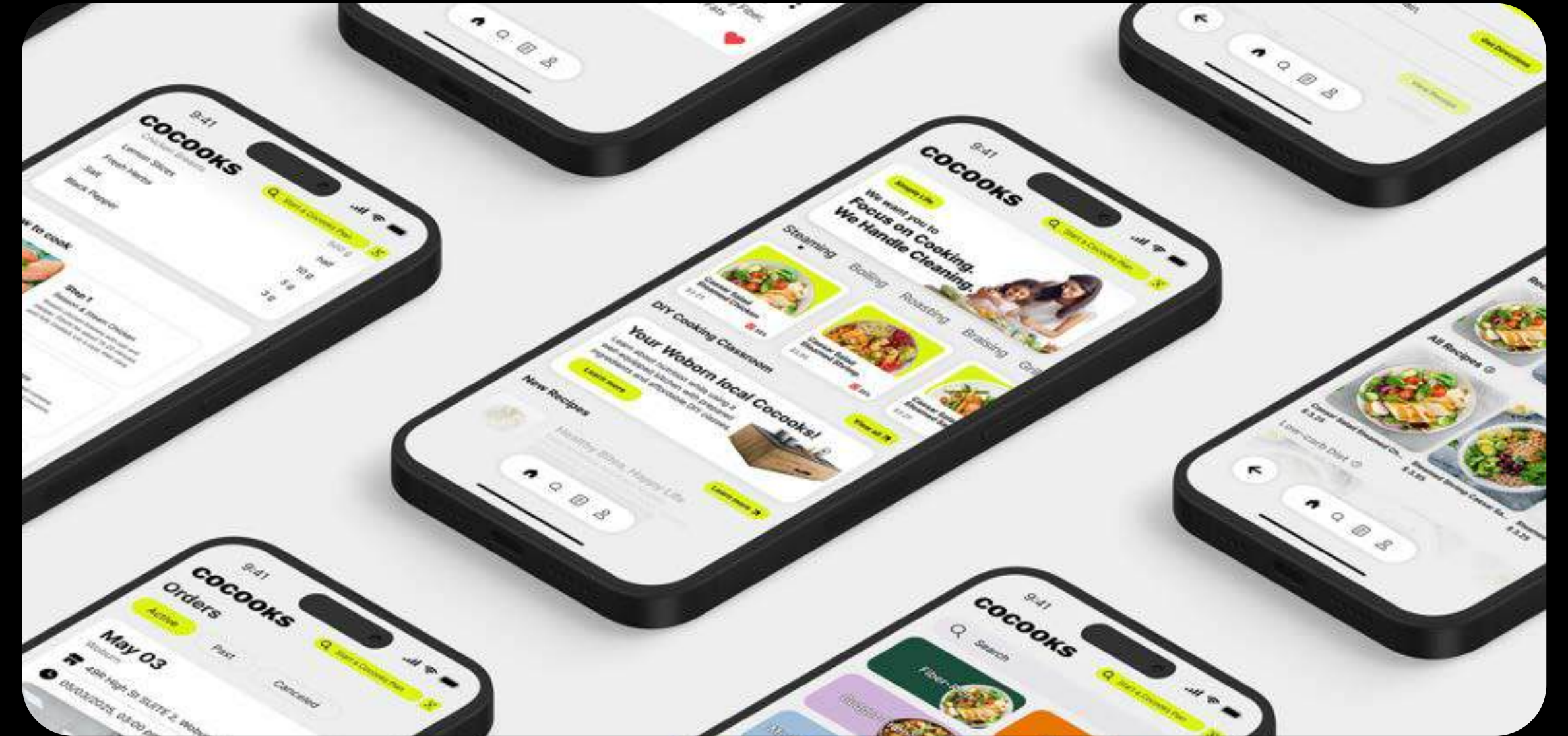
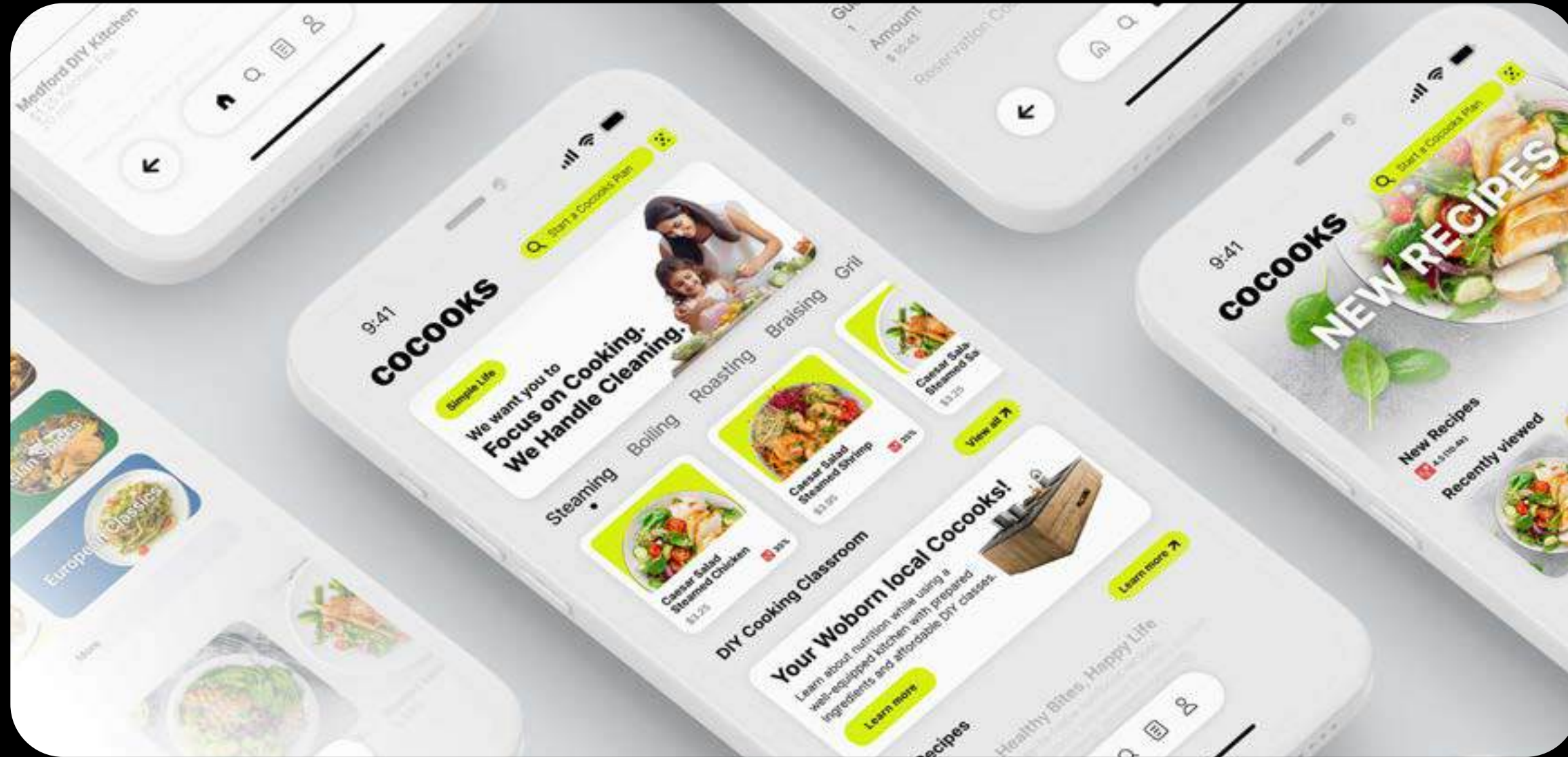


ILLUSTRATION SYSTEM LIST

- FRESH TOFU SOURCING
- PREMIUM BEEF SUPPLY
- LOCAL SEAFOOD & CRAB
- HEALTHY NOODLES
- POULTRY & FARMING
- FRESH CATCH FISH





- Bridging creative strategy with advanced technology at the MIT Media Lab, I engaged in diverse HCI research.

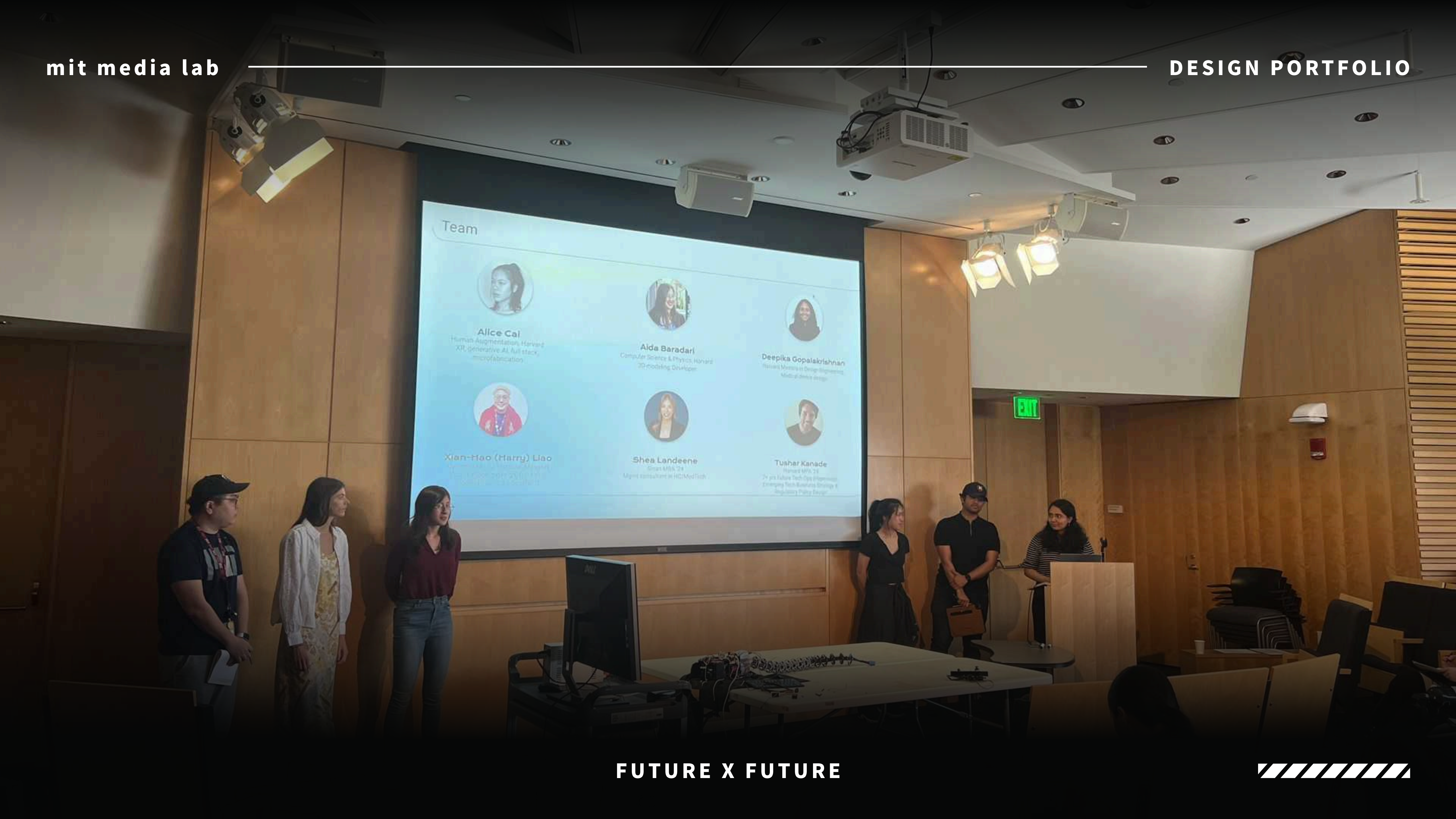
PROJECT SCOPE
PRODUCT DESIGN, USER TESTING, PROTOTYPING

mit media lab

Robotic Limb Project

We conducted 22 user tests to evaluate AR-robotic limb interactions. By identifying critical pain points through real-time feedback, we guided optimization for the next prototyping phase.





MIT Robotic Limb Case Study

AR Controlled Continuum Supernumerary Robotic Limb (ARCCSRL)
Alice Cai, Aida Baradari, Shea Landeene, Harry Liao, Deepika Gopalakrishnan, Tushar Kanade

Abstract

Supernumerary robotic limbs (SRLs) have the potential to improve human abilities by extending the range and number of actions that can be performed. However, the existing control interfaces for SRLs are low-level, requiring a high cognitive load for users. This research proposes a novel augmented reality (AR)-controlled continuum supernumerary robotic limb (ARCCSRL) that allows users to deliver high-level control commands using eye gaze input supported by secondary voice commands. To assess the effectiveness, usability, and limitations of this system, comparative user studies were conducted using a joystick-controlled continuum SRL and an AR-controlled continuum SRL. Given several technical difficulties during user testing, the results do not suggest that using AR as a control method leads to improved accuracy or efficiency in human-robot interaction, but does leave room for improvements in future studies to further validate the hypothesis of AR controls serving to improve accuracy and efficiency compared to traditional control mechanisms.

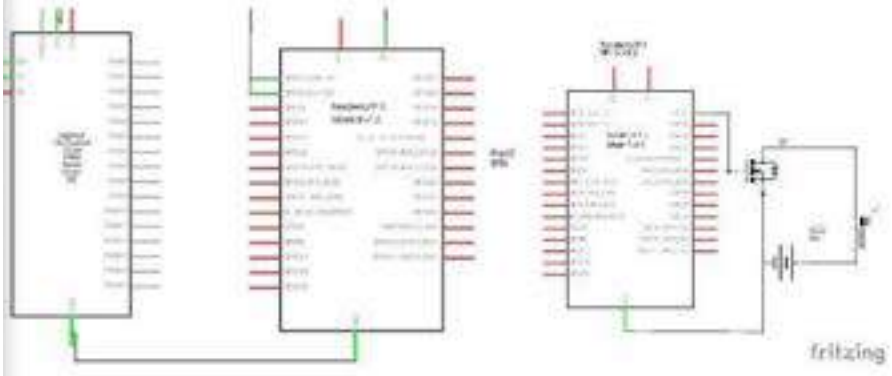
Keywords—supernumerary robotic limb, continuum robot, augmented reality, eye-gaze tracking, voice command recognition, embodiment

I. Introduction

Supernumerary robotic limbs (SRLs) are robotic appendages worn by human users [1]. SRLs have the potential to improve human abilities by extending the range and quantity of actions that can be performed, especially for those with disabilities [2]. However, the current control interfaces for SRLs are often low-level, which results in both a steep learning curve and a high cognitive load for the human user [1]. This can limit the effectiveness and usability of these devices. As such, there is a need for a more natural, intuitive, and user-friendly control system. Additionally, most SRLs are rigid robots designed similarly to human limbs. While this provides a familiar form factor, it limits the possible range of motions, applications, and uses for these devices. To improve the usability and effectiveness of SRLs, there is a need to explore alternative designs and control systems that leverage the potential of the technology to create more versatile and user-friendly limbs.

This research proposes a novel AR-controlled continuum supernumerary robotic limb (ARCCSRL) that allows users to deliver high-level control commands using eye gaze input supported by secondary voice commands. Continuum robots have shown greater flexibility and adaptability compared to rigid robots, making them suitable for integrating with the human body for a wider range of applications and environments [3]. In this phase of research, the control system focuses on allowing users to direct the SRL to move toward and grab objects in the operation space. This research contributes (1) a three-segment continuum SRL hardware design, (2) a machine learning-based proportional-integral-derivative (PID) robotic controller for the robotic limb, (3) a novel AR control language using eye gaze and voice command to identify target objects and desired operations, and (4) insights from human subject experimentation using the SRL. By using high-level commands and taking advantage of the computational capabilities of the machine in the human-machine system, the research aims to reduce the cognitive load and learning curve of controlling the SRL. Additionally, the proposed control system can be extended to off-body machines, allowing for more effective coordination and teamwork between humans and machines. It is hypothesized that compared to traditional control methods such as a joystick, the ARCCSRL is expected to demonstrate enhanced user performance and enhanced intuitive control via shorter average time to complete tasks, greater average accuracy in movement execution, and greater average rating in user experience.

[1] B. Yang, J. Huang, X. Chen, C. Xiong and Y. Hasegawa, "Supernumerary Robotic Limbs: A Review and Future Outlook," in IEEE Transactions on Medical Robotics and Biomim, vol. 3, no. 1, pp. 621-639, Aug. 2021, doi: 10.1109/TMRB.2021.3066016.
[2] Pratto-Luiza, Domenico, et al. "Human augmentation by wearable supernumerary robotic limbs: review and perspectives." Progress in Biomedical Engineering 3.4 (2021): 042005.
[3] Elena, Giannaccini Maria. "Novel design of a soft lightweight pneumatic continuum robot arm with decoupled variable stiffness and posttensioning." Soft robotics (2018).

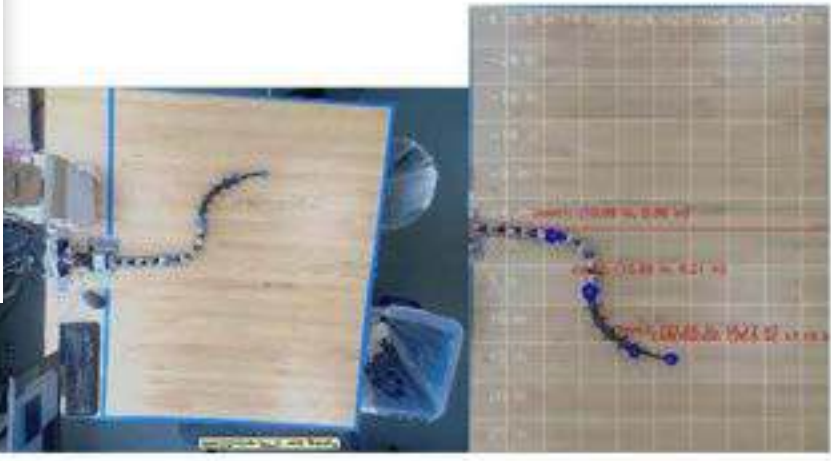


Left: Simple servo driver to Raspberry Pi connection, Right: Electromagnet actuation circuit diagram

Using a MOSFET and an additional, external 9V power supply, the electromagnet can be directly controlled from the Raspberry Pi.

Both of these circuits are used in the manual and AR controller, and are therefore generalizable. The manual controller uses both Joystick and buttons to control the servo motors and the electromagnet respectively, whereas the AR controller uses predicted positions and language to control servo motors and the electromagnet respectively.

Machine Learning Controller. To control the SRL, a machine learning model was developed to map the desired end effector position to the appropriate set of servo angles. Instead of a traditional kinematics-based controller, we trained a neural network to predict the needed servo angles for a desired location. We developed a data collection rig consisting of an overhead camera to collect video data of the limb configurations at different combinations of servo angles. We sampled a large parameter space, with over 800 data points and developed a computer-vision based post-processing system to level the alignment, segment the joints and end effectors, and map the pixel positions to real world dimensions.



The set of collected data shows an elliptic configuration space which we will be trying to predict with our neural network.

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Experimental Design

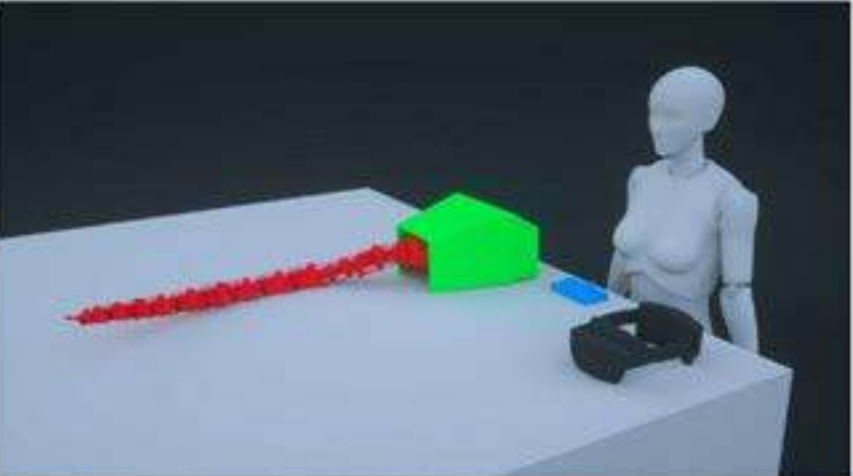


Fig 1. Continuum SRL user testing experimental design: Participants were asked to stand behind the table-mounted continuum SRL and use either a joystick controller or AR controller to execute the task of moving the SRL end effector to the center of a metal target, and activate the electromagnet at the tip of the SRL end effector to grab the metal target; participants were allotted 5 minutes of training with each controller system in which they could engage with the respective controller system freely and ask clarifying questions; Three trials were conducted for each controller type: Trial 1 included a metal target to the left of the SRL, Trial 2 included a metal target to the right of the SRL, Trial 3 included a metal target in the center in front of the SRL; Participants completed the series of trials in the same order (1-3), using the ending location of the previous trial as the origin point for the next trial; Accuracy was measured as the distance between the center of the SRL end effector and the center of the metal target; Efficiency was measured as the time to execute each task



Fig 2. AR control system. (Right) Unity development interface with calibration system. (Left) Researcher calibrating system by moving spherical table corner markers.



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Controller type	Trial	Avg. time to target (sec)	Avg. distance from target center (mm)	Total trials	Participants per trial (N)
AR	Trial1 (left)	21.8	21.3	20	20
	Trial2 (right)	15.8	28.7	17	17
	Trial3 (center)	15.5	28.8	18	18
	Overall	17.9	26.1	55	21
Joystick	Trial1 (left)	20.8	12.1	22	22
	Trial2 (right)	22.6	14.0	22	22
	Trial3 (center)	15.4	10.4	22	22
	Overall	19.6	12.2	66	22

Continuum SRL user testing results, summary (N_{AR-controller}=21; N_{joystick-controller}=22): Comparison of AR and joystick controlled SRL when completing three trials in which the user was instructed to use the joystick controller system to move the SRL end effector to touch the center of a metal target and activate the magnet at the tip of the SRL end effector; Green denotes statistically significant outperformance; Red denotes statistically significant underperformance; Grey denotes not statistically significant difference

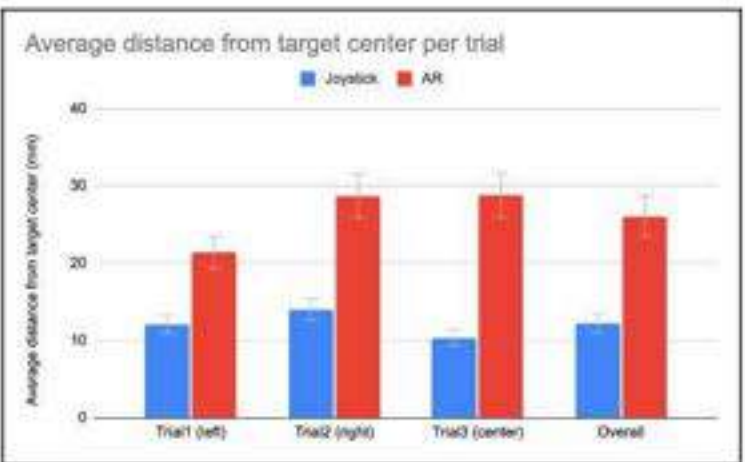


Fig 2. Continuum SRL user testing results, average accuracy trial distribution (N_{AR-controller}=21; N_{joystick-controller}=22):

15 A. Cai, A. Baradari, S. Landeene, H. Liao, D. Gopalakrishnan, T. Kanade

MIT Brain+AI Case Study

Your Brain on ChatGPT: Accumulation of Cognitive Debt when Using an AI Assistant for Essay Writing Task^Δ

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MIT Media Lab
Cambridge, MA

Eugene Hauptmann
MIT
Cambridge, MA

Ye Tong Yuan
Wellesley College
Wellesley, MA

Jessica Situ
MIT
Cambridge, MA

Xian-Hao Liao
Mass. College of Art
and Design (MassArt)
Boston, MA

Ashly Vivian Beresnitsky
MIT
Cambridge, MA

Iris Braunstein
MIT
Cambridge, MA

Pattie Maes
MIT Media Lab
Cambridge, MA

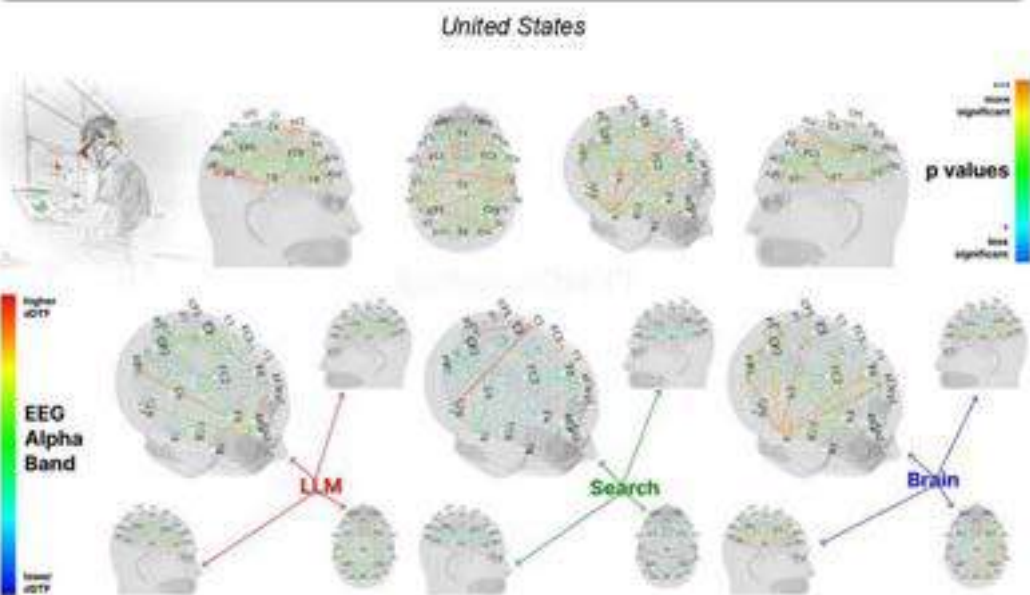


Figure 1. The dynamic Direct Transfer Function (ddTF) EEG analysis of Alpha Band for groups: LLM, Search Engine, Brain-only, including p-values to show significance from moderately significant (*) to highly significant (***).

¹ Nataliya Kosmyna is the corresponding author, please contact her at nkosmyna@mit.edu
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with ChatGPT or similar LLM tools. The examples of the questions included: 'How often do you use LLM tools like ChatGPT?', 'What tasks do you use LLM tools for?', etc.

The total time required to complete stage 1 of the experiment was approximately 15 minutes.

Stage 2: Setup of the Enobio headset

All participants regardless of their group assignment were then equipped with the Neuroelectrics Enobio 32 headset, [128], used to collect EEG signals of the participants throughout the full duration of the study and for each session (Figure 4). The sampling rate of the headset was 500 Hz. Ground and reference were on an ear clip, with reference on the front and ground on the back. Each of 32 electrode sites had hair parted to reveal the scalp and Spectra 360 salt- and fluoride-free electrode gel was placed in Ag/AgCl wells, at each location. EEG channels were visually inspected at the start of each session after setup. Each participant was asked to perform eyes closed/eyes open task, blinks, and a jaw clench to test the response of the headset.

The experimenter then requested that participants turn off and isolate their cell phones, smartwatches, and other devices in the bin to isolate them from the participants during the study.

Once the headset was turned on, participants were informed about the movement artifacts and were asked not to move unnecessarily during the session. Then the Neuroelectrics® Instrument controller (NIC2) application and the BioSignal Recorder application were turned on. The NIC2 application is provided by Neuroelectrics and used to record EEG data. The BioSignal application was used to record a calibration test (Stage 3). All recordings and data collection were performed using The Apple MacBook Pro.

The total time required to complete stage 2 of the experiment was approximately 25 minutes.



Figure 4. Participant during the session, while wearing Enobio headset, AttentionU headset, using BioSignal recorder software.

choice", followed by "increased choice" and "power uncertainty". The Search Engine group had "homeless person" and "moral obligation". See Figure 25 below.

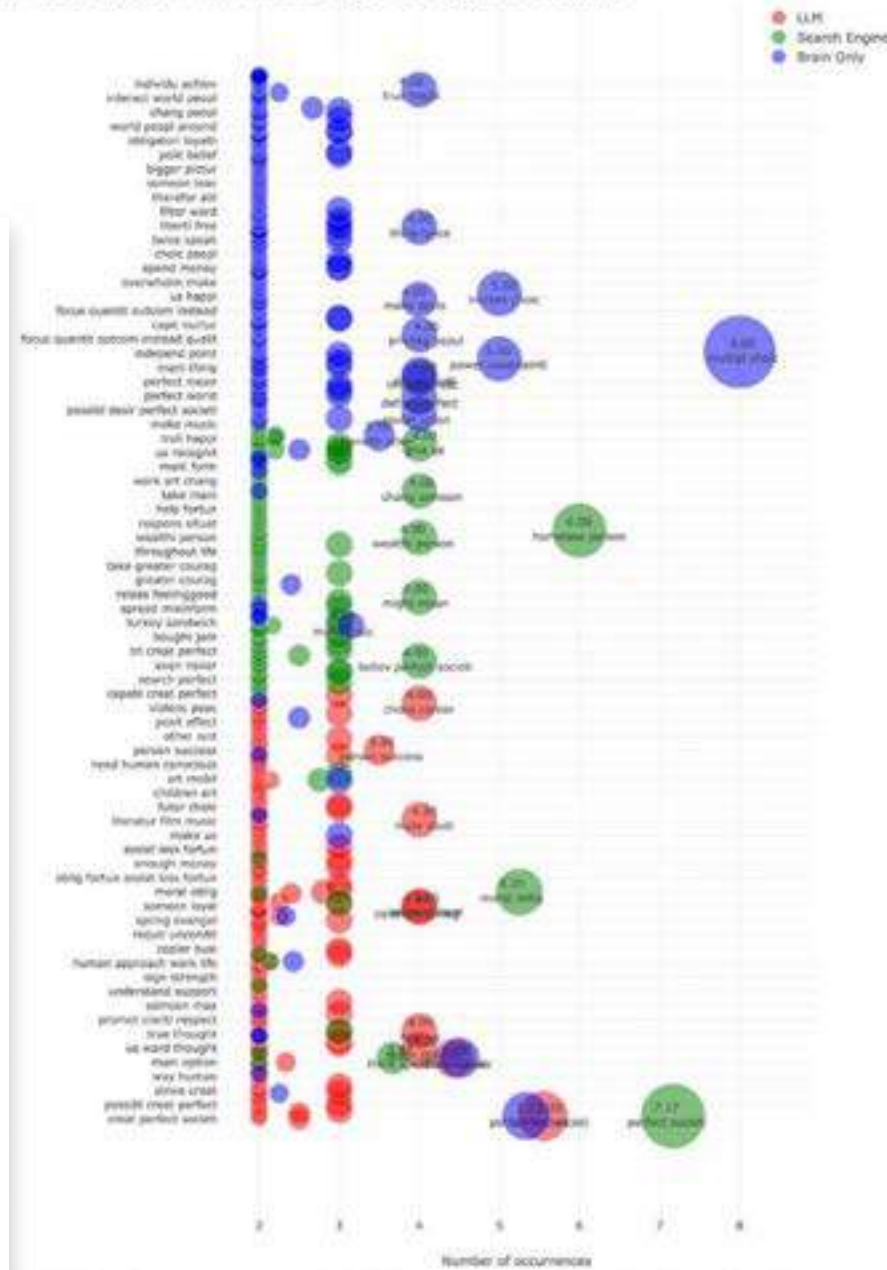
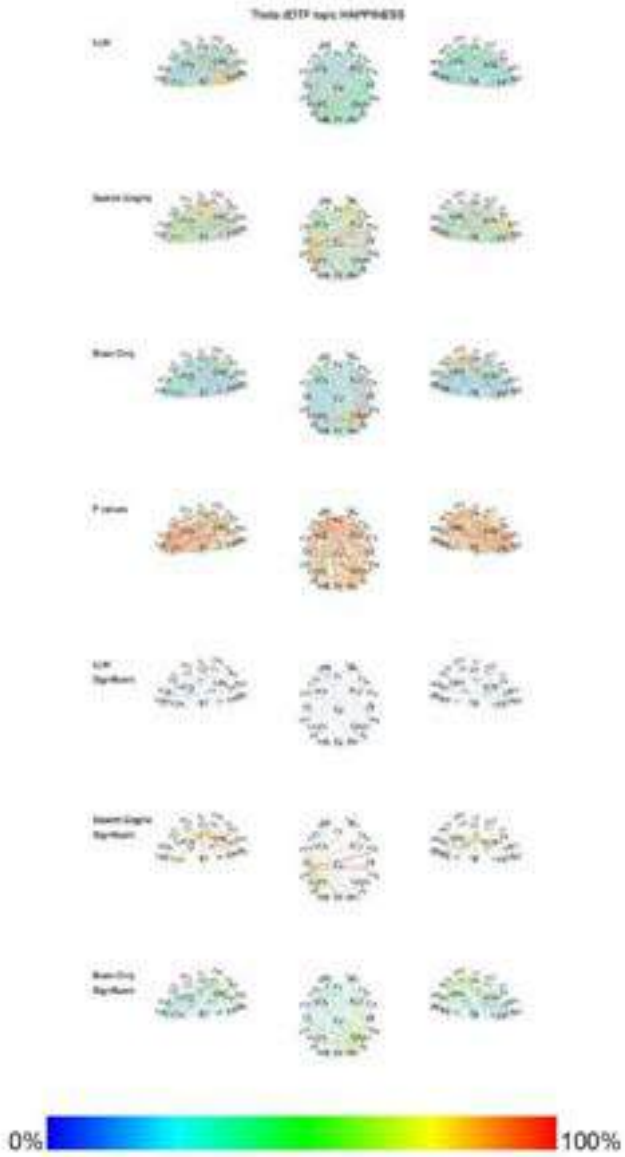


Figure 25. Total n-grams used across the topics per group. This figure displays a distribution of n-grams aggregated across all topics with each radius representing the frequency of the n-gram used within the topic. X-axis shows most frequent n-grams. Y-axis shows frequency of n-grams within the essays.

Figure F3, F3), supports the hypothesis that the tool generated much of the language, and the participants showed little integration or reflection.

Further, the neural and linguistic evidence points toward a more externally scaffolded writing process with minimal reliance on endogenous semantic or affective regulation, potentially diminishing the influence of tool-driven composition over self-generated reflection.



Her, Way, Home

- Transforming traditional concert spaces into dynamic multimedia journeys through synchronized digital landscapes.

PROJECT SCOPE
LIVE VISUAL DESIGN, INTERACTIVE MEDIA

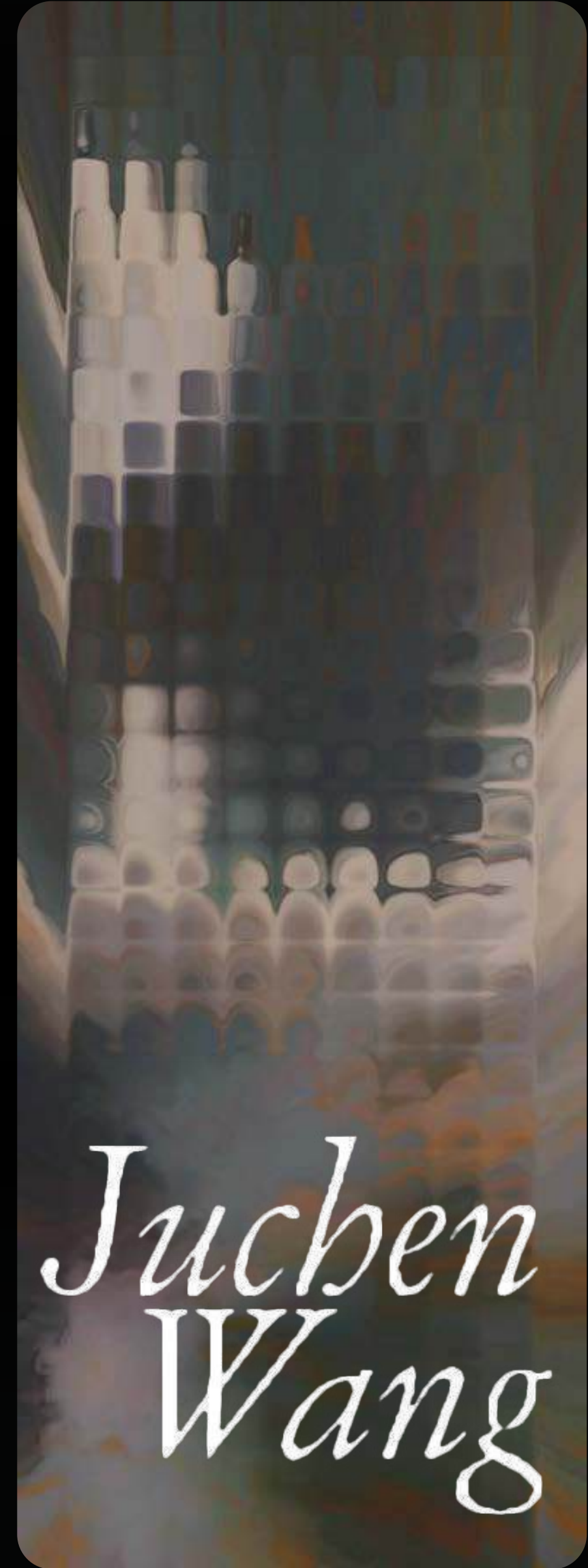
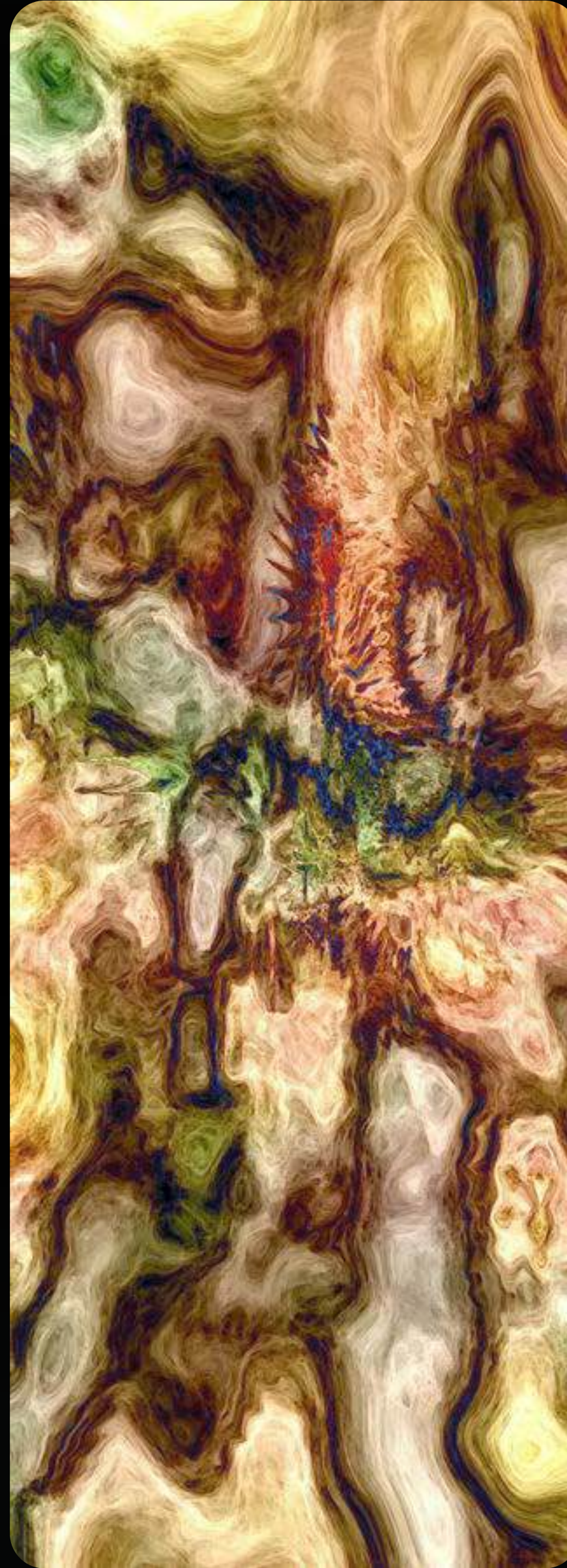
Her Way Home Recital

Live Visual Design & Projection

Her Way Home is an interdisciplinary collaborative project with a pianist from the New England Conservatory. It aims to integrate visual arts with classical music performance to create an immersive audio-visual experience.



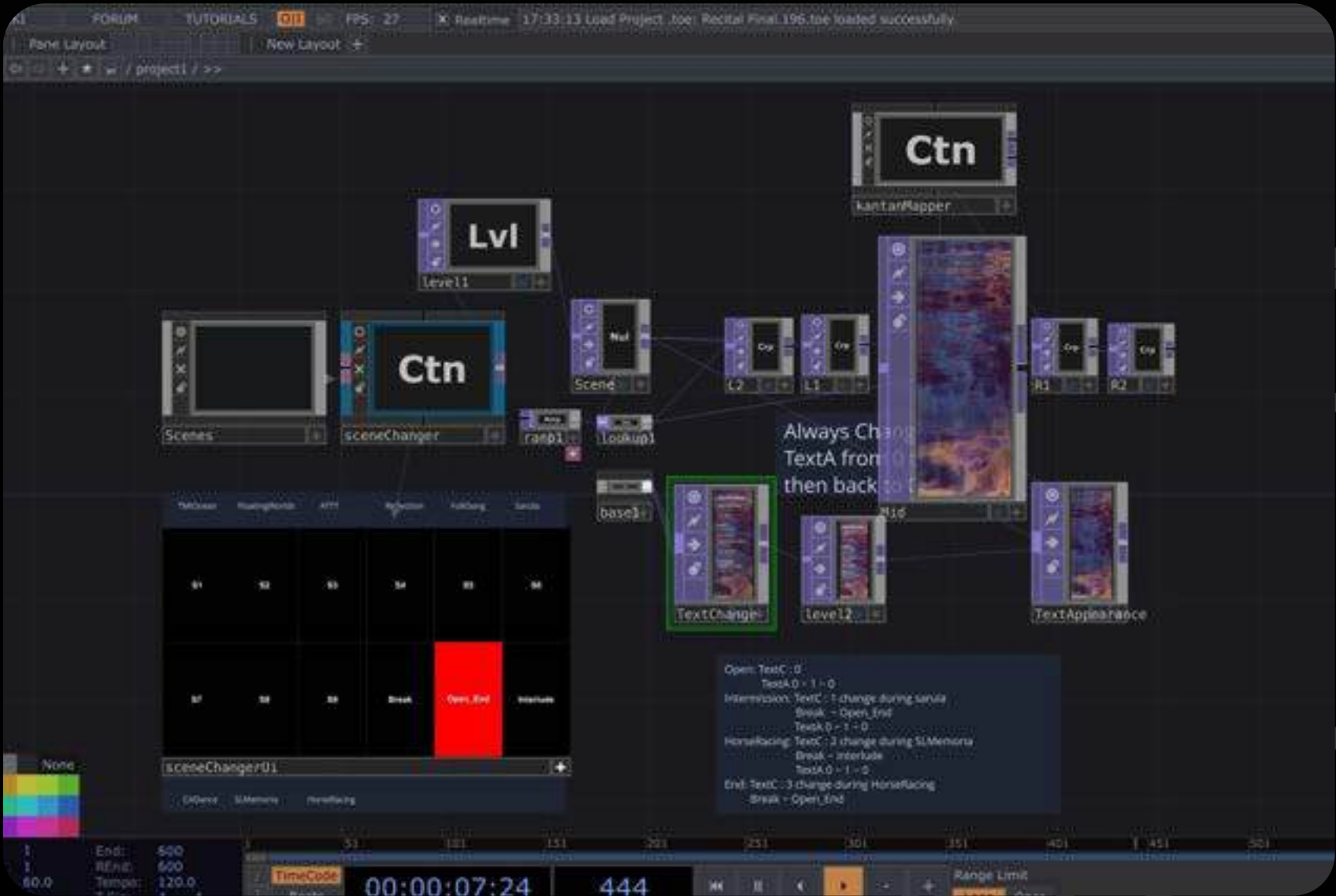
*Juchan
Wang*



Her Way Home Recital

Live Visual Design & Projection

Collaborated with artist Juchen Wang to design and execute real-time generative visuals for a live recital at the New England Conservatory. By integrating dynamic projection mapping with the musical performance, we created an innovative art format that elevated the audience's immersive auditory and visual experience.



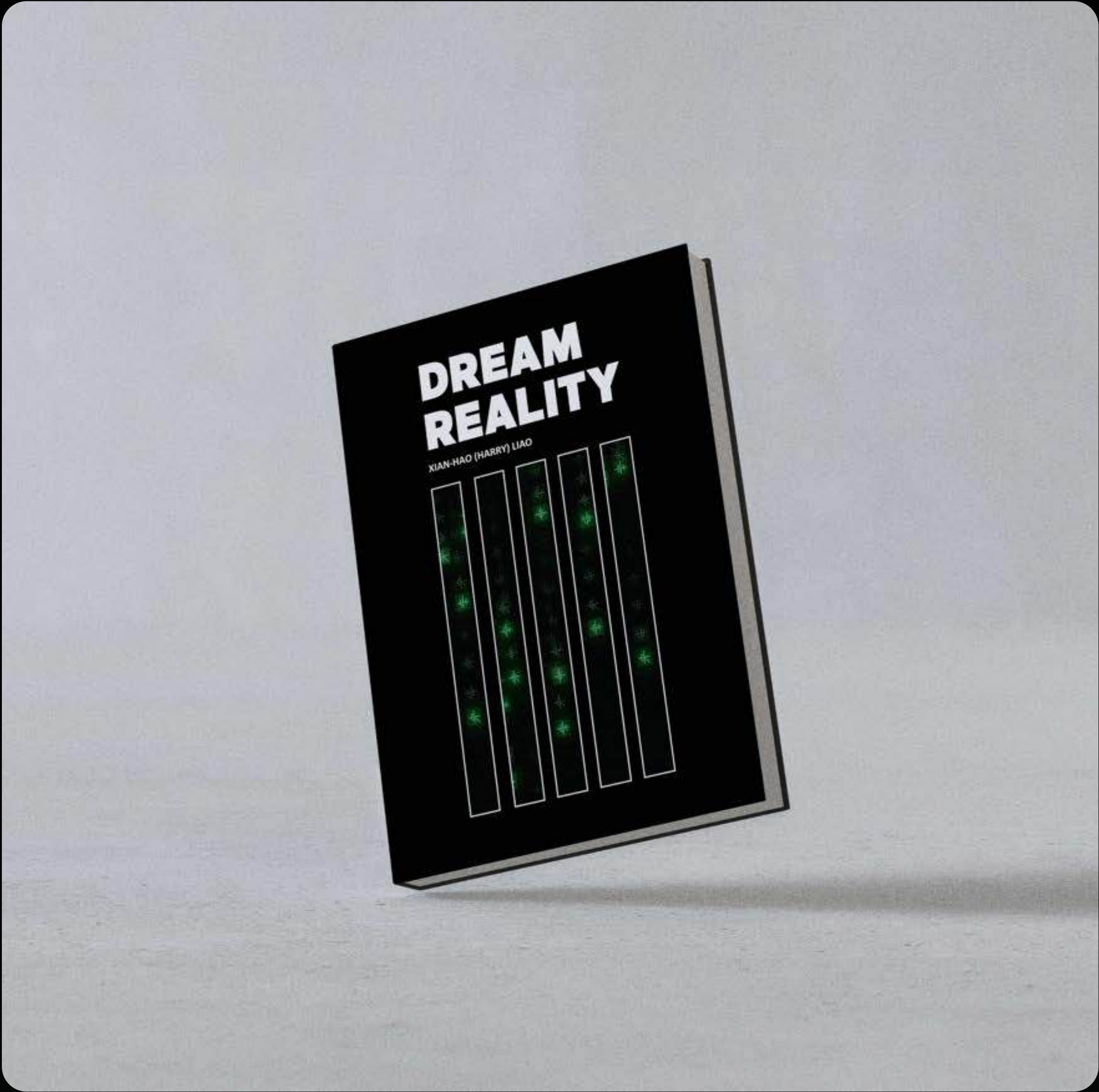
PROJECT TYPE
LIVE VISUAL DESIGN / INTERACTIVE MEDIA

DREAM REALITY



- Investigated the intersection of dream, memory, and perception through an immersive master's thesis, employing deep user research and an interactive hardware system.

PROJECT SCOPE
THESIS RESEARCH, EXPERIENCE DESIGN, HARDWARE PROTOTYPING



01

**Lucid Dreams
History Exhibition**



02

**Dream
Handcrank**



03

**Dive into
the Dream**



04

**Aware of Dreams
in Dreaming**



Thesis Book Design



BJ's

Deals
you can
crush.



[BJs.com/simple-savings](https://www.bjs.com/simple-savings)

- Led the large-scale visual design for the BJ's Nationals Park OOH Ads campaign in Washington DC during the summer internship.

PROJECT SCOPE
OOH CAMPAIGN, MARKETING MATERIAL DESIGN



Don't let your grocery savings get away.



Now open in La Vergne.



1-year
The Club Card Membership

Join for \$55*
and get a
\$40

✓ Unbeatable savings:
Save up to 25% off grocery
store prices every day.

✓ Risk free:
Try BJ's risk free, with our
100% money-back
membership satisfaction
guarantee.



MDP Summer Internship Final

Harry Liao / MDP Summer Intern - Creative Team
Master of Fine Arts: Dynamic Media Institute / MassArt
Social & Behavioral Research Investigator / MIT Media Lab



- The project focused on modernizing the MassArt DMI's digital presence through a contemporary visual language and improved information architecture.

PROJECT SCOPE
UI/UX DESIGN / WEBSITE REDESIGN / BRANDING

DMI Institute Website

Website Redesign

Role: Designer / Developer

Industry: Education

Date: 2024

Institute Website Upgrade

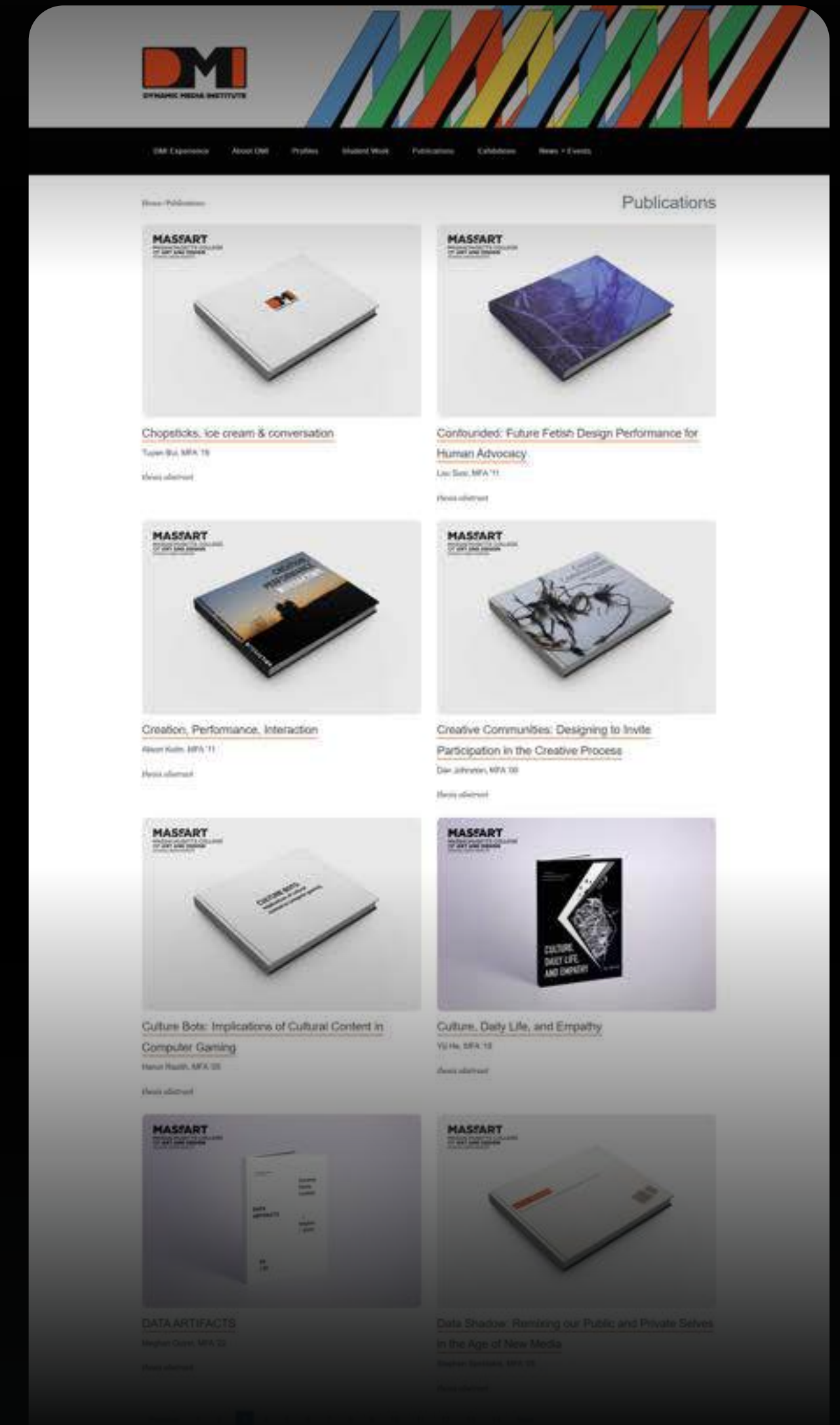
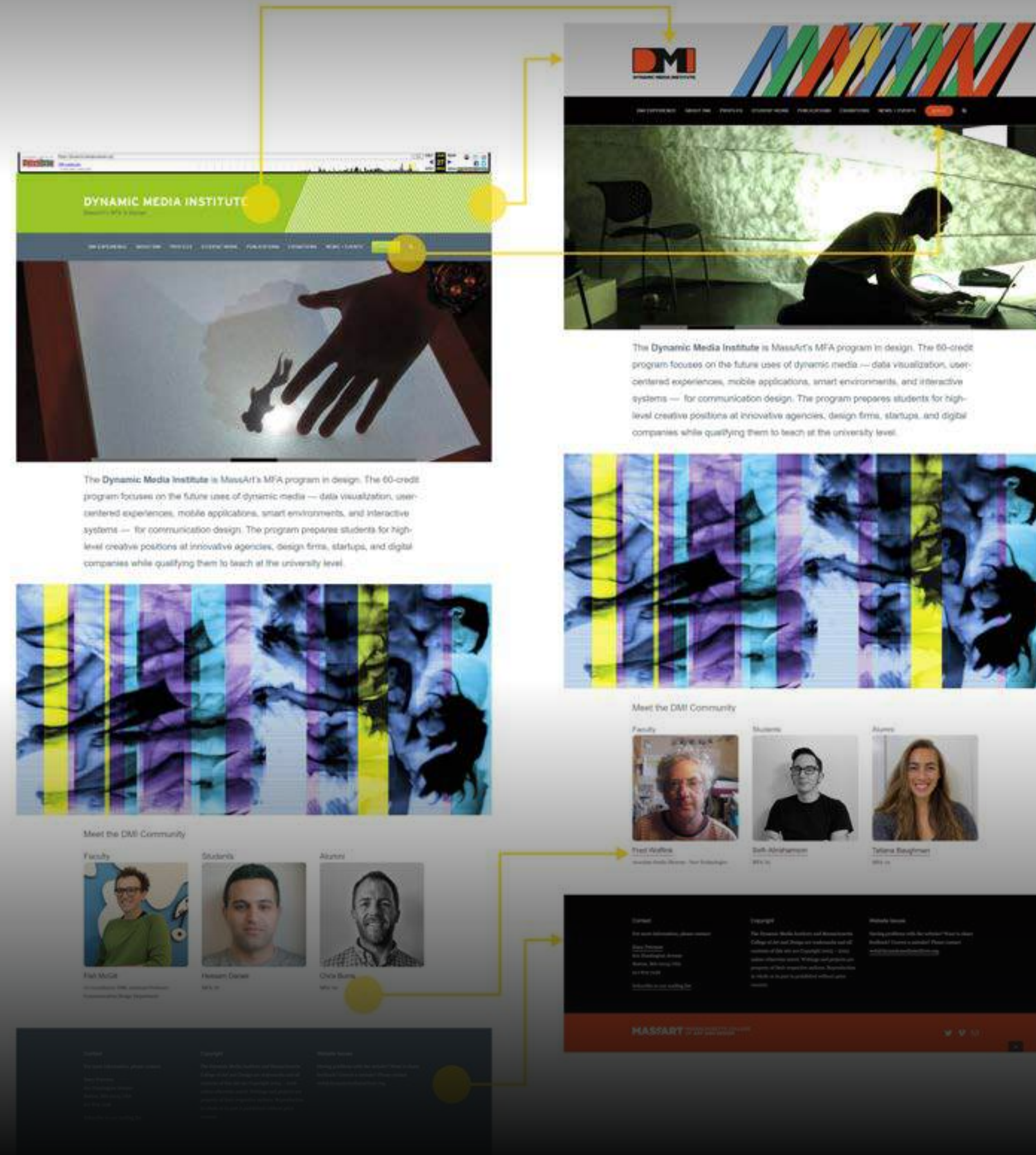
The Dynamic Media Institute (DMI) at MassArt required a visual and functional overhaul to reflect its cutting-edge design curriculum.

Focused on improving user navigation and simplifying the back-end interface for faculty and student content management.

Redesigned layout features a bold, grid-based aesthetic that aligns with the institute's modernist brand identity.

PROJECT TYPE

UI/UX DESIGN / WEBSITE REDESIGN / BRANDING



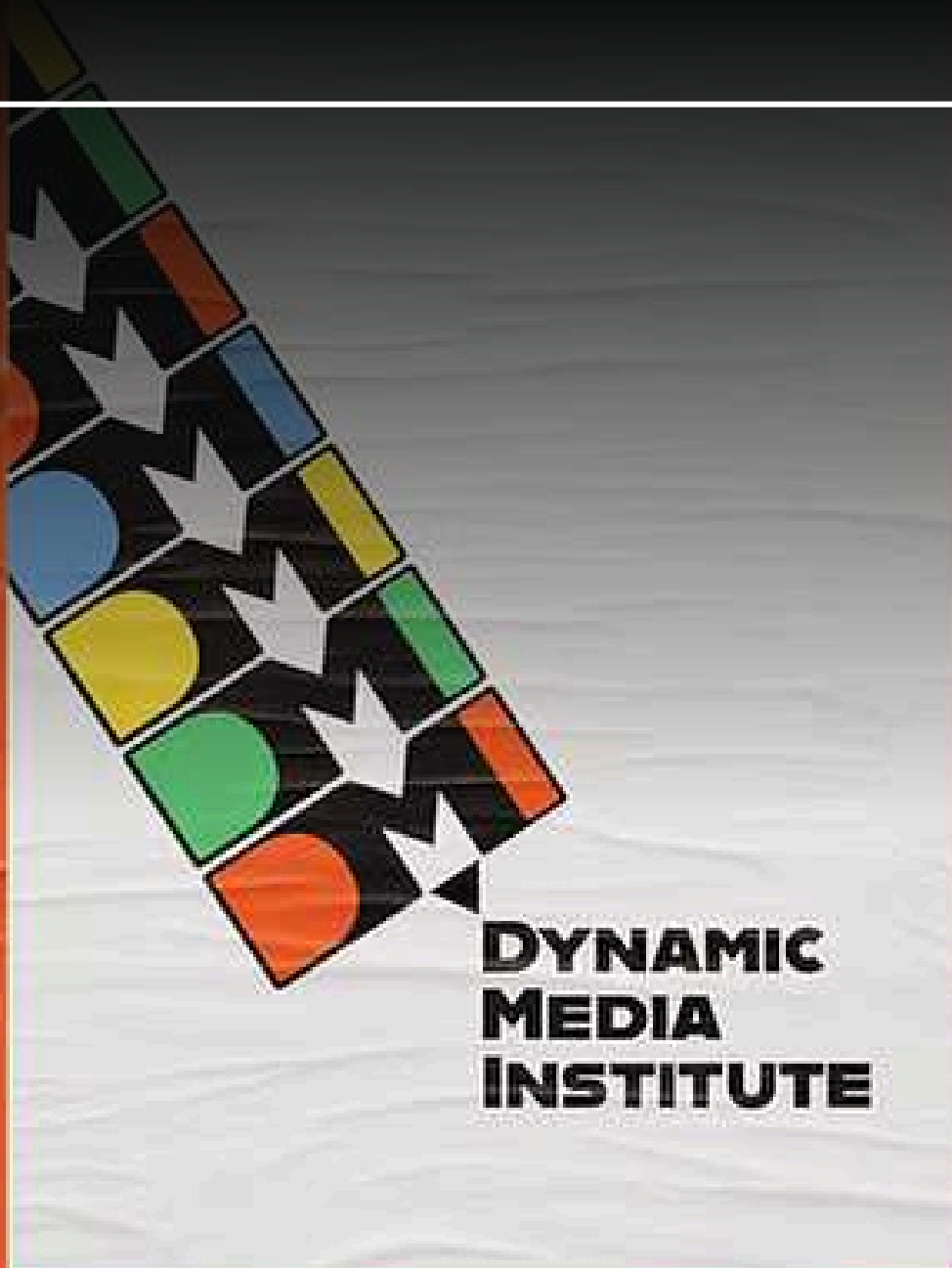
Institute
communication to
community, education,
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perspectives.
of dynamic and
experimenting



Dynamic Media Institute

DMI is ever-evolving, always seeking communication to
new frontiers. DMI is known for its community, education,
experimentation, and versatility. Based on this, we have
developed a system that will evolve and expand as the
institute introduces new initiatives and perspectives.

We welcome the opportunity to build and develop our
powerful brand with you, collaborating and experimenting
together as we take DMI into the future.



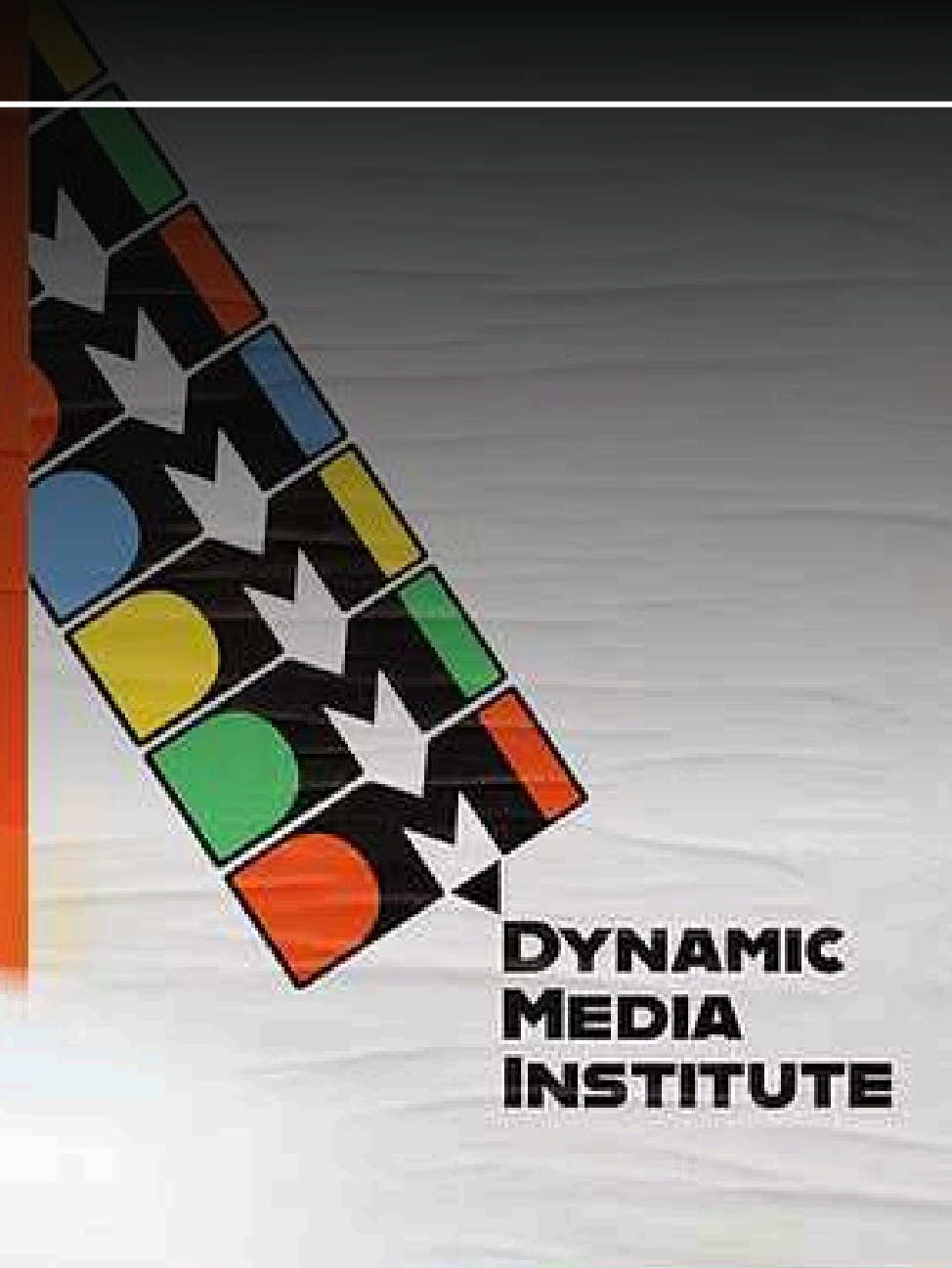
**DYNAMIC
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INSTITUTE**



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**DYNAMIC
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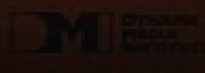
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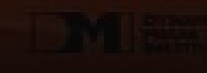
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Institute



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**DY
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1



ABC DESIGN

1

1

代言人 王陽明



1

- Involved creating cohesive social media graphics, marketing materials, and promotional short videos to elevate brand presence and engage the target audience.

PROJECT SCOPE
VISUAL DESIGN / SOCIAL MEDIA MARKETING / VIDEO EDITING



Brand Campaign

Visual Content & Video Editing

Executed visual strategy and content design for the Sunny Wang ambassador campaign.

A sophisticated and modern visual language was designed to communicate the premium positioning of the high-end stroller line.

Cohesive social media graphics and dynamic short-form videos were created to maximize digital impact.

The campaign successfully elevated the brand's premium image and drove significant engagement.

PROJECT TYPE
SOCIAL MEDIA MARKETING





S P O R T

von Sternplatz



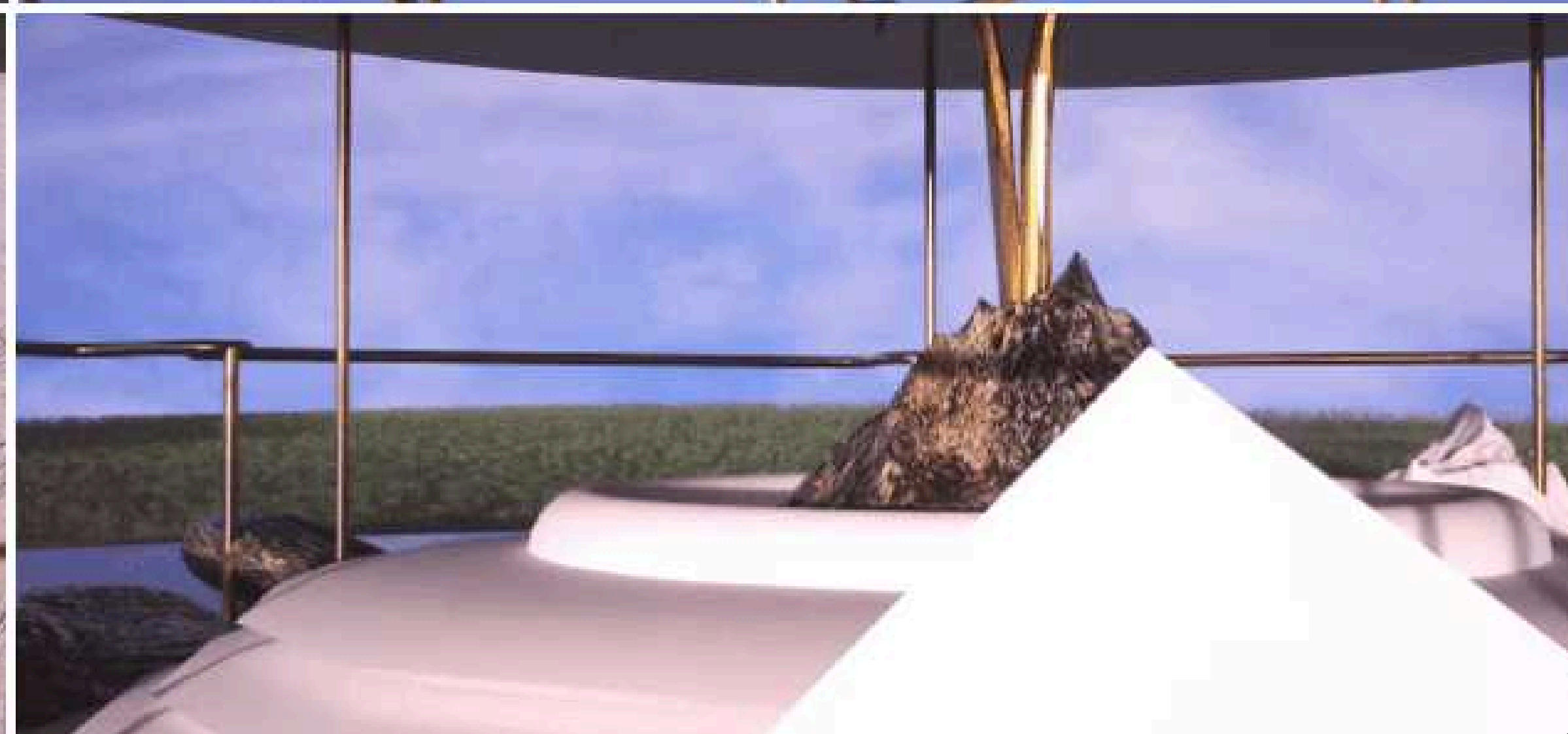
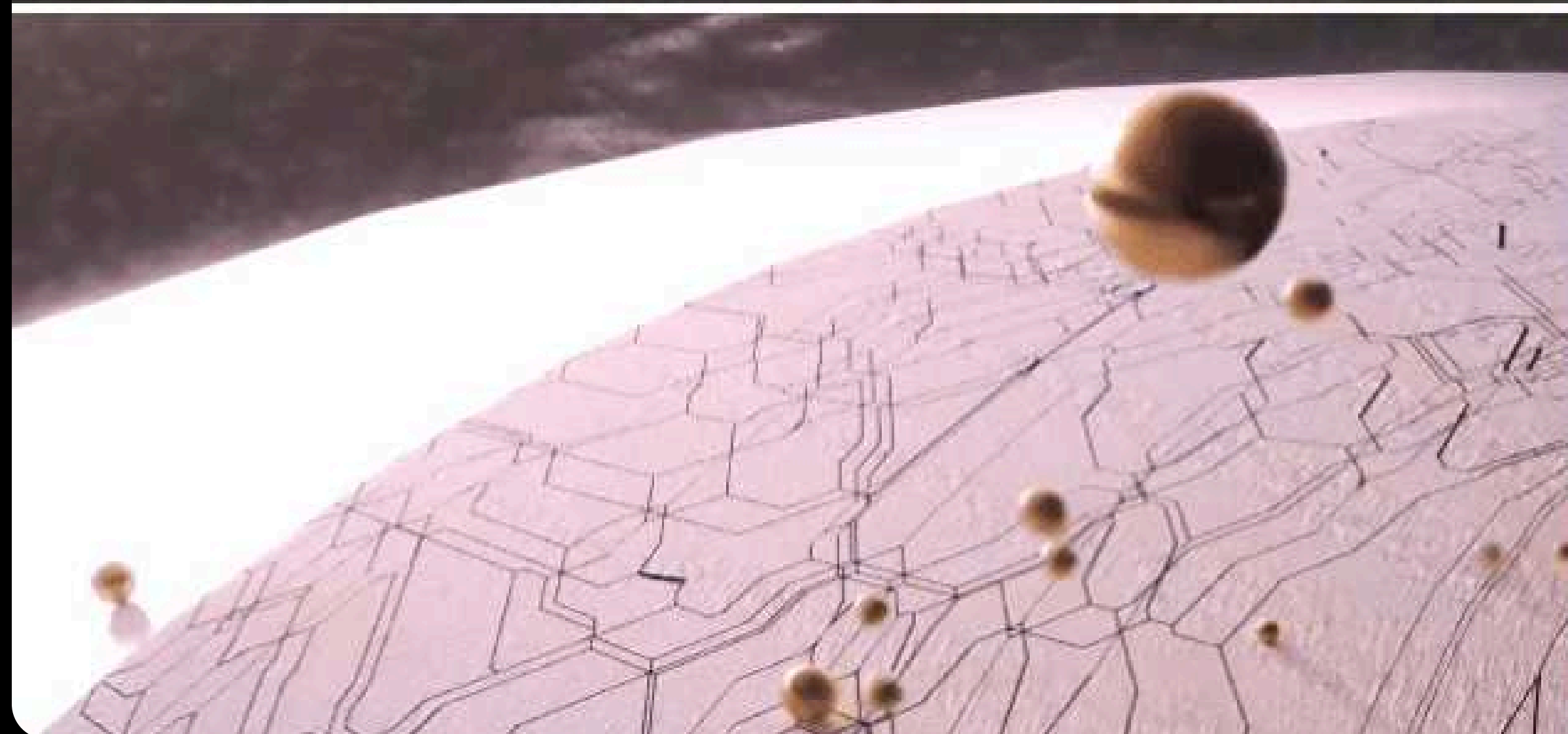
von Sternplatz

FOR-U



- Merging creative strategy with visual storytelling to build a cohesive music video and brand experience.

PROJECT SCOPE
VISUAL STORYTELLING, ART DIRECTION, MV PRODUCING



FOR-U

DESIGN PORTFOLIO

FOR-U

RENO GORDON

FT. LUNA

A LYRICS VIDEO BY BROTHERHOOD CASUAL

DIRECTED BY RNGDN

3D MODELING.YANG PO ZHI

EDIT&DI.RNGDN

MANAGEMENT. BEAR TIM

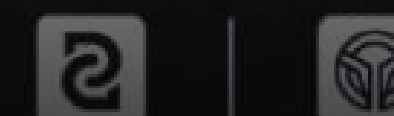
LINE PRODUCER

TW

RENO GORDON,XUN XIE

LA

AUSTIN LEEDS



BROTHERHOOD
CASUAL

FOR-U

3D Animation & MV Design

I directed and produced the 3D animated music video, set in a futuristic oasis that merges nature with technology. The project encompassed end-to-end execution, including 3D visual animation, original music production, and comprehensive brand identity design, delivering a cohesive and immersive audiovisual experience.



PROJECT TYPE

3D ANIMATION / MUSIC PRODUCTION / MV DESIGN



- An exclusive, integrated platform combining portfolio showcase for student entrepreneurs.

C-CUBE

UI/UX Design & Information Architecture

I structured and designed a comprehensive website for the C-CUBE Incubation Base.

By systematically categorizing their diverse services into six core sections, from entrepreneurship counseling to design exhibitions.

We provides an intuitive and seamless user experience that connects student entrepreneurs with vital resources.

創業輔導



優質活動



設計展售



創業課程



開發設備



團隊進駐



模擬圖片



網頁介面設計



1. 首頁



2. 創業課程



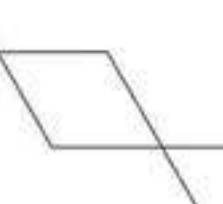
3. 優質活動



4. 設計展售



5. 開發設備



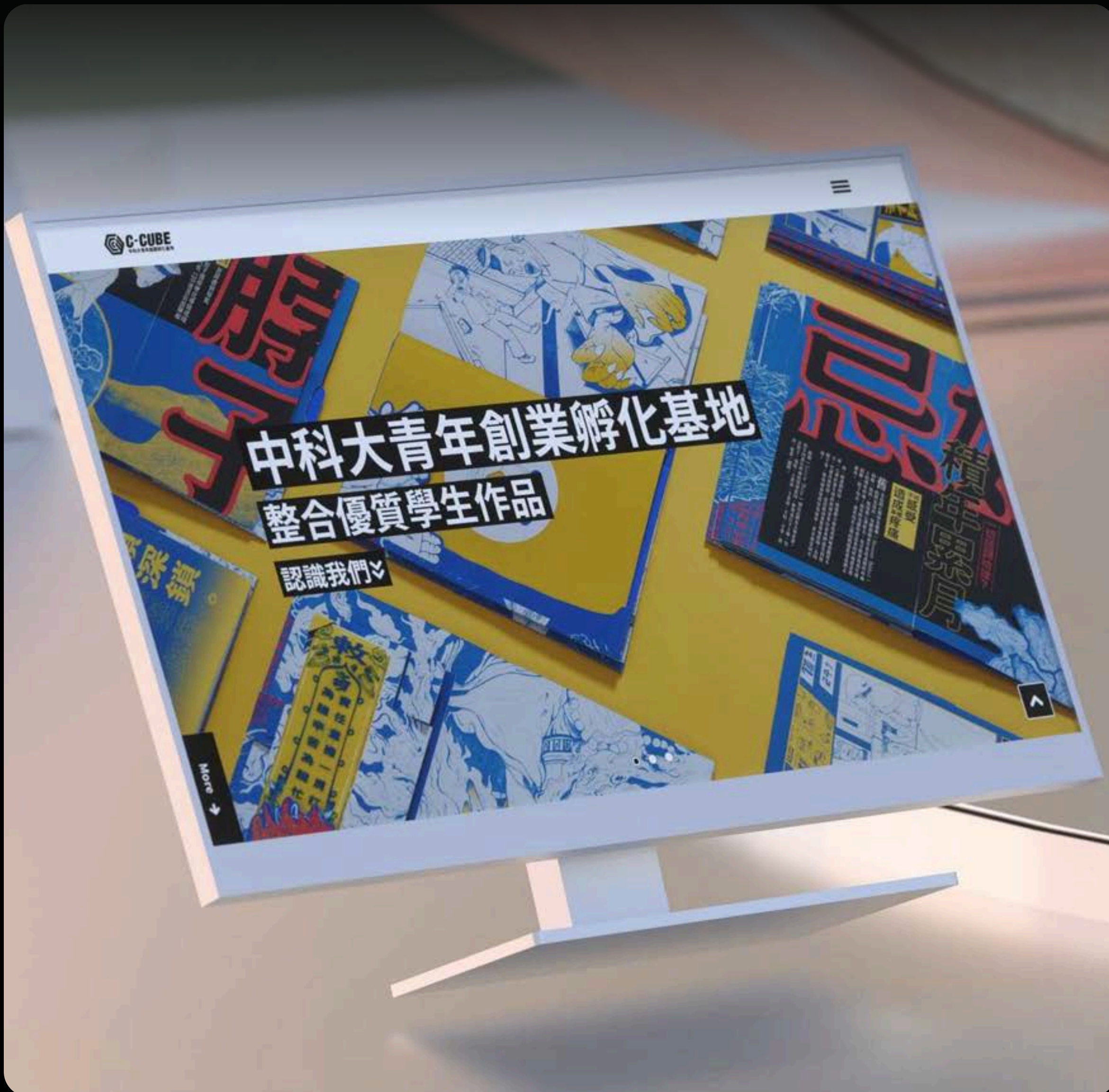
6大頁面

創業課程、優質活動、設計展售
開發設備、創業輔導、團隊進駐

PROJECT TYPE

UI/UX DESIGN / WEB DESIGN



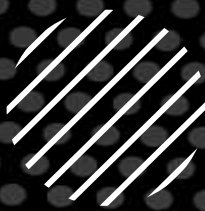




XIAN-HAO (HARRY) LIAO



THANK YOU



DESIGN PORTFOLIO

